Unit 7. Generalised Linear Models I: Categorical Responses

1 Binary Responses

FREES §11.1; ISLR §4.1–4.2 (PA reading list, not SRM). SRM learning outcomes 2(a), 2(c).

Remark (The classical assumptions; Unit 2)**.** A1–A5 are Unit 2's classical assumptions on $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ with $\mathbf{X}$ fixed: linearity (A1), zero-mean errors (A2), $\text{var}(\boldsymbol{\varepsilon}) = \sigma^2\mathbf{I}_n$ with $\sigma^2 > 0$ (A3), $\text{rank}(\mathbf{X}) = k+1$, with $n > k+1$ for inference (A4), and normal errors (A5).

1.1 Binary variables

Definition 7.1 (Binary response; FREES §11.1)**.** A **binary** response takes the values 0 and 1. With $\pi_i = \mathsf{P}(y_i = 1)$ the response y_i has the **Bernoulli** distribution with parameter π_i. Throughout, $y_1, \ldots, y_n$ are independent and $\mathbf{x}_i = (1, x_{i1}, \ldots, x_{ik})'$ are fixed, the rows of an $n \times (k+1)$ matrix $\mathbf{X}$.

Proposition 7.2 (Bernoulli moments; FREES §11.1)**.** If y is Bernoulli with parameter π, then $\mathsf{E}\, y = \pi$ and $\text{var}\, y = \pi(1-\pi)$, and π determines the distribution.

Proof. $\mathsf{E}\, y = 0 \cdot (1-\pi) + 1 \cdot \pi$, and $y^2 = y$ gives $\mathsf{E}\, y^2 = \pi$, so $\operatorname{var} y = \pi - \pi^2$. The two probabilities are π and $1-\pi$. □

Example 7.1 (Hospitalisation by sex; FREES §11.1, Table 11.1)**.** FREES's MEPS sample: $n = 2000$ adults aged 18 to 65, with `POSEXP` $= 1$ when inpatient expenditure is positive, that is, when the person was hospitalised. The 33 persons of race `OTHER` are merged into `WHITE`, which reproduces FREES's tables, whose race variable has four levels:

```
H <- read.csv("data/frees/csv/HealthExpend.csv")
H$POSEXP <- 1 * (H$EXPENDIP > 0)
table(H$RACE); H$RACE[H$RACE == "OTHER"] <- "WHITE"
```

```
##
## ASIAN BLACK NATIV OTHER WHITE
##    86   295    22    33  1564
```

```
H$RACE <- relevel(factor(H$RACE), ref = "WHITE"); H$REGION <-
 ↪  relevel(factor(H$REGION), ref = "WEST")
H$EDUC <- relevel(factor(H$EDUC), ref = "LHIGHSC"); H$PHSTAT <-
 ↪  relevel(factor(H$PHSTAT), ref = "EXCE")
```

```r
H$INCOME <- relevel(factor(H$INCOME), ref = "POOR")
tab_sex <- table(POSEXP = H$POSEXP, GENDER = H$GENDER); tab_sex
```

```
##       GENDER
## POSEXP   0   1
##      0 902 941
##      1  44 113
```

```r
round(100 * prop.table(tab_sex, 2), 1)
```

```
##       GENDER
## POSEXP    0    1
##      0 95.3 89.3
##      1  4.7 10.7
```

Of the 946 men (**GENDER** = 0), 44 were hospitalised, 4.7%; of the 1054 women, 113, 10.7%.

1.2 The linear probability model

Definition 7.3 (Linear probability model; FREES §11.1)**.** The **linear probability model** is $y_i = \mathbf{x}_i'\boldsymbol{\beta} + \varepsilon_i$ with $\mathsf{E}\,\varepsilon_i = 0$, so that $\pi_i = \mathbf{x}_i'\boldsymbol{\beta}$, and it is fitted by least squares with $\operatorname{rank}(\mathbf{X}) = k+1$.

Proposition 7.4 (Three drawbacks; FREES §11.1)**.** In the linear probability model with every π_i in $(0, 1)$:

(i) **fitted values can be poor:** if $\mathbf{x}'\boldsymbol{\beta} \in [0, 1]$ for every $\mathbf{x} = (1, x_1, \ldots, x_k)'$ with $x_1, \ldots, x_k \in \mathbb{R}$, then $\beta_1 = \cdots = \beta_k = 0$; and a fitted value $\hat{y}_i = \mathbf{x}_i'\hat{\boldsymbol{\beta}}$ outside $[0, 1]$ makes $\hat{y}_i(1 - \hat{y}_i)$, the variance the fit implies, negative;

(ii) **heteroscedasticity:** $\operatorname{var}\varepsilon_i = \pi_i(1 - \pi_i)$, so A3 holds only if every π_i is π or $1 - \pi$ for one π;

(iii) **residual analysis is meaningless:** ε_i takes the two values $1 - \pi_i$ and $-\pi_i$, so A5 fails, and the residuals $e_i = y_i - \hat{y}_i$ lie on the two lines $e = 1 - \hat{y}$ and $e = -\hat{y}$.

Proof. (i) If $\beta_j \neq 0$ for some $j \geqslant 1$, the vector $\mathbf{x}$ with $x_j = t$ and the other slopes' coordinates 0 gives $\mathbf{x}'\boldsymbol{\beta} = \beta_0 + t\beta_j$, unbounded in t. And $t(1 - t) < 0$ exactly when $t < 0$ or $t > 1$.

(ii) $\varepsilon_i = y_i - \pi_i$ has the variance of y_i, which is $\pi_i(1-\pi_i)$ by Proposition 7.2; and $t(1-t) = s(1-s)$ iff $(t-s)(1-t-s) = 0$.
(iii) y_i is 1 or 0, so ε_i is $1-\pi_i$ or $-\pi_i$ and e_i is $1-\hat{y}_i$ or $-\hat{y}_i$. A normal variable has positive probability on every interval, and a variable with two values does not. □

Remark (FREES §11.1)**.** Two remedies address (ii). Two-stage weighted least squares applies Unit 4's theorem **weighted least squares** with the weights $1/\{\hat{y}_i(1-\hat{y}_i)\}$ of a first least squares fit in place of the unknown $1/\{\pi_i(1-\pi_i)\}$, and needs every $\hat{y}_i$ in $(0,1)$. Least squares with robust standard errors needs nothing of the fitted values: as $|\varepsilon_i| \leqslant 1$, the fourth moments are bounded and Unit 4's theorem **consistency of the robust standard errors** applies whenever its design conditions hold.

Example 7.2 (The linear probability model on the MEPS data)**.** Least squares of `POSEXP` on the explanatory variables of FREES's reduced model of §11.4 (sex, race, region, education, self-rated physical health, activity limitation, income and insurance), with HC0 standard errors:

```
f_red <- POSEXP ~ GENDER + RACE + REGION + EDUC + PHSTAT + ANYLIMIT + INCOME +
  insure
```

```
lp <- lm(f_red, data = H); fv <- fitted(lp); el <- residuals(lp)
round(c(min_fitted = min(fv), max_fitted = max(fv)), 5); c(below_0 = sum(fv < 0),
 ↪  above_1 = sum(fv > 1))
```

```
## min_fitted max_fitted
##   -0.06903    0.43648
## below_0 above_1
##     157       0
```

```
Xl <- model.matrix(lp); Bl <- solve(crossprod(Xl))
se_hc0 <- sqrt(diag(Bl %*% crossprod(Xl * el) %*% Bl))
round(cbind(estimate = coef(lp), se_ols = coef(summary(lp))[, 2], se_hc0 =
 ↪  se_hc0)[c("GENDER", "PHSTATPOOR", "insure"), ], 4)
```

```
##            estimate se_ols se_hc0
## GENDER       0.0454 0.0119 0.0114
## PHSTATPOOR   0.2452 0.0345 0.0593
## insure       0.0652 0.0153 0.0136
```

```
op <- par(mar = c(3.2, 3.8, 0.6, 0.6), mgp = c(2.1, 0.6, 0))
plot(fv, el, pch = 16, cex = 0.35, xlab = "fitted value", ylab = "residual");
 ↪  abline(v = 0, lty = 3); par(op)
```

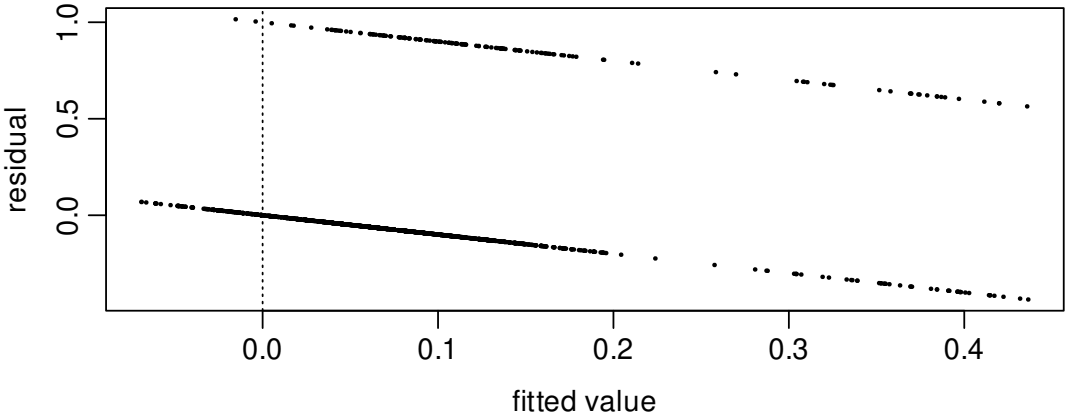


Figure 7.1: Residuals against fitted values of the linear probability model for the MEPS data: the two lines $e = 1 - \hat{y}$ and $e = -\hat{y}$ of Proposition 7.4, with fitted values left of the dotted line below 0.

The fitted values run from -0.06903 to 0.43648, and 157 are negative: the variances of Proposition 7.4(i) are negative there, and two-stage weighted least squares cannot be formed. The residuals lie on the two lines of (iii). The robust standard error of `PHSTATPOOR` is 0.0593, against 0.0345 from least squares.

Proposition 7.5 (Coding a binary response; ISLR §4.2)**.** In least squares with an intercept, replacing every y_i by $1-y_i$ replaces every fitted value $\hat{y}_i$ by $1-\hat{y}_i$, so the classification "1 if $\hat{y}_i > 1/2$" is the same under either coding, observations with $\hat{y}_i = 1/2$ excepted; and $n^{-1}\sum_i \hat{y}_i = \bar{y}$, the proportion of ones.

Proof. $\hat{\mathbf{Y}} = \mathbf{HY}$ with $\mathbf{H}$ the hat matrix of Unit 2, the orthogonal projection onto $\mathcal{C}(\mathbf{X})$, which contains $\mathbf{1}$; so $\mathbf{H1} = \mathbf{1}$ and $\mathbf{H}(\mathbf{1}-\mathbf{Y}) = \mathbf{1}-\hat{\mathbf{Y}}$. And $\hat{y}_i > 1/2$ iff $1-\hat{y}_i < 1/2$. Finally $\mathbf{1}'\hat{\mathbf{Y}} = \mathbf{1}'\mathbf{HY} = (\mathbf{H1})'\mathbf{Y} = \mathbf{1}'\mathbf{Y}$, $\mathbf{H}$ being symmetric. □

Example 7.3 (Coding a three-level response; ISLR §4.2)**.** A response with three unordered levels has no such invariance. `origin` in `Auto` is American, European or Japanese; least squares of a numerical code on `mpg`, `displacement`, `weight` and `year`, each fitted value classified to the level with the nearest code, under two codings:

```
cl_or <- c("American", "European", "Japanese")[Auto$origin]
code_fit <- function(code) {
  f <- fitted(lm(code[cl_or] ~ mpg + displacement + weight + year, data = Auto))
  names(code)[sapply(f, function(v) which.min(abs(code - v)))] }
```

```
c1 <- code_fit(c(American = 1, European = 2, Japanese = 3))
c2 <- code_fit(c(Japanese = 1, American = 2, European = 3))
round(c(disagree = mean(c1 != c2), accuracy_first = mean(c1 == cl_or),
        accuracy_second = mean(c2 == cl_or)), 3)
```

```
##        disagree  accuracy_first accuracy_second
##           0.577           0.589           0.622
```

The two codings classify 57.7% of the 392 cars differently, with accuracies 58.9% and 62.2%. A numerical code imposes an order and equal spacing on the levels, and a different code is a different linear model.

Example 7.4 (Credit card default: linear against logistic; ISLR §4.1–4.2, §4.3.1, Figure 4.2)**.** ISLR's simulated **`Default`**: 10,000 credit card holders, with **`default`** **`Yes`** or **`No`**, monthly **`balance`**, annual **`income`** and **`student`** status. Least squares and the logit on **`balance`**:

```
Df <- Default; Df$y <- 1 * (Df$default == "Yes"); lpD <- lm(y ~ balance, data = Df)
lgD <- glm(default ~ balance, family = binomial, data = Df)
round(c(default_rate = mean(Df$y), negative_fitted_lpm = mean(fitted(lpD) < 0),
        min_fitted_lpm = min(fitted(lpD)), mean_fitted_lpm = mean(fitted(lpD)),
        mean_fitted_logit = mean(fitted(lgD))), 4)
```

```
##        default_rate negative_fitted_lpm      min_fitted_lpm
##              0.0333               0.3123             -0.0752
##     mean_fitted_lpm   mean_fitted_logit
##              0.0333               0.0333
```

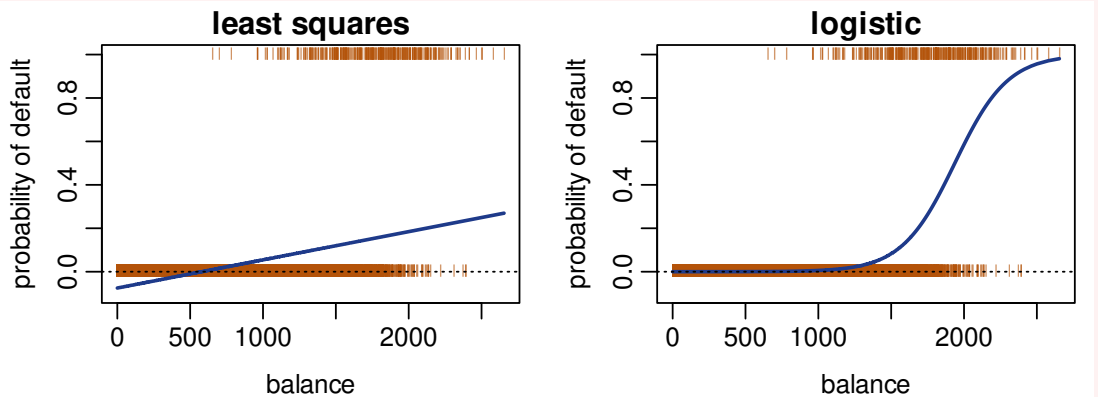


Figure 7.2: `Default`: fitted probabilities of default against `balance` by least squares (left) and by the logit (right), with the responses as ticks at 0 and 1 (ISLR Figure 4.2).

The default rate is 3.33%. Least squares gives negative fitted probabilities to 31.2% of the card

holders, down to −0.0752, the first drawback of Proposition 7.4; the logit's lie in (0, 1). The least squares fitted values average 0.0333, the default rate, by Proposition 7.5.

2 Logistic and Probit Regression

FREES §11.2; ISLR §4.3.1 (PA). SRM learning outcomes 2(a), 2(c).

2.1 Response functions

Definition 7.6 (Binary regression; FREES §11.2.1)**.** Let $\pi(\cdot)$ be a continuous, strictly increasing distribution function on $\mathbb{R}$. The **binary regression** with response function π has $\pi_i = \mathsf{P}(y_i = 1) = \pi(\mathbf{x}_i'\boldsymbol{\beta})$, equivalently $\pi^{-1}(\pi_i) = \mathbf{x}_i'\boldsymbol{\beta}$, and π^{-1} is the **link function**. The two cases used here are the **logit**, $\pi(z) = \Lambda(z) = 1/(1+e^{-z}) = e^z/(1+e^z)$, and the **probit**, $\pi(z) = \Phi(z)$, the standard normal distribution function. The choice $\pi(z) = z$ is the linear probability model, which is not a distribution function.

Definition 7.7 (Odds and logit; FREES §11.2.4, ISLR (4.3)–(4.4))**.** For $p \in (0,1)$ the **odds** are $p/(1-p)$, and the **logit** is $\text{logit}(p) = \ln\{p/(1-p)\}$, the log odds.

Proposition 7.8 (The logistic function; FREES §11.2.4)**.** Λ is a continuous, strictly increasing distribution function, symmetric in the sense $\Lambda(-z) = 1 - \Lambda(z)$, with $\Lambda' = \Lambda(1 - \Lambda)$ and $\operatorname{logit} \Lambda(z) = z$. So the logit model is $\operatorname{logit} \pi_i = \mathbf{x}_i'\boldsymbol{\beta}$.

Proof. $\Lambda'(z) = e^{-z}/(1 + e^{-z})^2 > 0$, and $\Lambda(z) \to 0$ and 1 as $z \to -\infty$ and $+\infty$. Multiplying numerator and denominator of $\Lambda(-z) = 1/(1 + e^z)$ by e^{-z} gives $e^{-z}/(1 + e^{-z}) = 1 - \Lambda(z)$, and the product of $\Lambda(z) = 1/(1 + e^{-z})$ and $1 - \Lambda(z)$ is $\Lambda'(z)$. Finally $\Lambda(z)/\{1 - \Lambda(z)\} = e^z$. □

Remark (FREES §11.2.4)**.** At odds against of $(1-p)/p$ to 1, a unit stake on an event of probability p wins $(1 - p)/p$ with probability p and is lost with probability $1 - p$, an expected gain of 0: odds are the payoffs of a fair bet.

2.2 Threshold and random utility

Theorem 7.9 (Threshold interpretation; FREES §11.2.2)**.** Let $y_i^* = \mathbf{x}_i'\boldsymbol{\beta} + \varepsilon_i^*$, where ε_i^* has a continuous distribution function G with $G(-a) = 1 - G(a)$ for all a, and let $y_i = \mathbb{1}\{y_i^* > 0\}$. Then $\mathsf{P}(y_i = 1) = G(\mathbf{x}_i'\boldsymbol{\beta})$. If $\varepsilon_i^* = \sigma\varepsilon_i$ with $\sigma > 0$ and ε_i has such a distribution function π, then

$\mathsf{P}(y_i = 1) = \pi(\mathbf{x}_i'\boldsymbol{\beta}/\sigma)$: the logit when ε_i is logistic and the probit when it is standard normal, with $\boldsymbol{\beta}$ and σ entering only through $\boldsymbol{\beta}/\sigma$.

Proof. $\mathsf{P}(y_i = 1) = \mathsf{P}(\varepsilon_i^* > -\mathbf{x}_i'\boldsymbol{\beta}) = 1 - G(-\mathbf{x}_i'\boldsymbol{\beta}) = G(\mathbf{x}_i'\boldsymbol{\beta})$. With $\varepsilon_i^* = \sigma\varepsilon_i$, $G(a) = \mathsf{P}(\varepsilon_i \leqslant a/\sigma) = \pi(a/\sigma)$, which inherits the symmetry. Λ is symmetric by Proposition 7.8, and Φ because the normal density is even. □

Remark. Multiplying $\boldsymbol{\beta}$ and σ by the same constant leaves every probability unchanged, so the data cannot identify the scale of the latent variable, and $\sigma = 1$ is a normalisation.

Proposition 7.10 (Random utility; FREES §11.2.3)**.** Let $\varepsilon_{i1}, \varepsilon_{i2}$ be independent with the extreme-value (Gumbel) distribution function $\mathsf{P}(\varepsilon \leqslant a) = \exp(-e^{-a})$. Then $\varepsilon_{i2} - \varepsilon_{i1}$ has distribution function Λ. So if the utilities $U_{ij} = u_i(V_{ij} + \varepsilon_{ij})$, with u_i strictly increasing, $V_{i1} = \mathbf{x}_i'\boldsymbol{\beta}$ and $V_{i2} = 0$, and $y_i = 1$ when $U_{i1} > U_{i2}$, then $\mathsf{P}(y_i = 1) = \Lambda(\mathbf{x}_i'\boldsymbol{\beta})$.

Proof. With $\gamma(a) = e^{-a}\exp(-e^{-a})$ the Gumbel density, independence and Fubini's theorem give

$$\mathsf{P}(\varepsilon_{i2} - \varepsilon_{i1} \leqslant t) = \int_{-\infty}^{\infty} \exp\{-e^{-(t+a)}\}\,\gamma(a)\,\mathrm{d}a = \int_0^{\infty} \exp\{-u(1+e^{-t})\}\,\mathrm{d}u = \frac{1}{1+e^{-t}},$$

substituting $u = e^{-a}$. As u_i is strictly increasing, $U_{i1} > U_{i2}$ iff $\varepsilon_{i2} - \varepsilon_{i1} < V_{i1} - V_{i2} = \mathbf{x}_i'\boldsymbol{\beta}$, an event of probability $\Lambda(\mathbf{x}_i'\boldsymbol{\beta})$, Λ being continuous. □

2.3 Interpreting the coefficients

Theorem 7.11 (Odds ratios and marginal effects; FREES §11.2.4, ISLR §4.3.1)**.** In the logit model, with the other explanatory variables fixed:

(i) if $x_{ij} \in \{0, 1\}$, then e^{β_j} is the **odds ratio**, the odds of $y_i = 1$ at $x_{ij} = 1$ divided by the odds at $x_{ij} = 0$;

(ii) $\beta_j = \partial \ln \mathrm{odds}(y_i = 1)/\partial x_{ij}$, so a unit increase of x_{ij} multiplies the odds by e^{β_j};

(iii) $\partial\pi_i/\partial x_{ij} = \beta_j\pi_i(1-\pi_i)$, largest in size at $\pi_i = 1/2$; in the probit model $\partial\pi_i/\partial x_{ij} = \beta_j\varphi(\mathbf{x}_i'\boldsymbol{\beta})$, with φ the standard normal density.

Proof. By Proposition 7.8, $\ln \text{odds}(y_i = 1) = \mathbf{x}_i'\boldsymbol{\beta}$. For (i) subtract this identity at $x_{ij} = 0$ from the identity at $x_{ij} = 1$: the left sides differ by the logarithm of the odds ratio and the right sides by β_j. For (ii) differentiate it, and note that a unit increase adds β_j to the log odds. For (iii) the chain rule gives $\beta_j \Lambda'(\mathbf{x}_i'\boldsymbol{\beta}) = \beta_j \pi_i (1 - \pi_i)$ by Proposition 7.8, and $t(1-t)$ is largest at $t = 1/2$; for the probit, $\Phi' = \varphi$. □

Remark (FREES §11.2.4)**.** β_j is the proportional rate of change of the odds, what FREES calls the proportional change in the odds ratio; an elasticity in the economists' sense would be $\partial \ln \text{odds} / \partial \ln x_{ij} = \beta_j x_{ij}$. An odds ratio is not a ratio of probabilities: p_1/p_0 and $\{p_1/(1-p_1)\}/\{p_0/(1-p_0)\}$ agree only when $p_1 = p_0$ or in the limit of small probabilities.

Example 7.5 (Odds of hospitalisation; FREES §11.2.4)**.** The odds of Example 7.1, from the counts and from FREES's rounded percentages:

```
p_w <- tab_sex["1", "1"] / sum(tab_sex[, "1"]); p_m <- tab_sex["1", "0"] /
  sum(tab_sex[, "0"])
round(c(odds_women = p_w / (1 - p_w), odds_men = p_m / (1 - p_m),
```

```
        odds_ratio = (p_w / (1 - p_w)) / (p_m / (1 - p_m)), probability_ratio = p_w
        ↪  / p_m), 4)
```

```
##        odds_women          odds_men        odds_ratio
##            0.1201            0.0488            2.4617
## probability_ratio
##            2.3050
```

```
round(c(odds_women = 0.107 / 0.893, odds_men = 0.047 / 0.953, odds_ratio = (0.107 /
↪  0.893) / (0.047 / 0.953),
        ratio_of_rounded_odds = 0.120 / 0.0493), 4)
```

```
##            odds_women              odds_men
##                0.1198                0.0493
##            odds_ratio ratio_of_rounded_odds
##                2.4296                2.4341
```

From the counts, the odds are 0.1201 for women and 0.0488 for men, an odds ratio of 2.4617, while the probabilities are in the ratio 2.3050: women's odds are 2.4617 times men's, their probability 2.3050 times. FREES's odds 0.120 and 0.0493 come from the rounded percentages, 0.1198 and 0.0493 here, and its odds ratio 2.434 is the ratio 2.4341 of those rounded odds.

2.4 Logit against probit

Proposition 7.12 (Logit against probit; FREES §11.2.1)**.** Λ and Φ are symmetric about 0, with value $1/2$ there.

(i) $\Lambda(\beta_L z)$ and $\Phi(\beta_P z)$ have the same slope at $z = 0$ iff $\beta_L = 4\beta_P/\sqrt{2\pi}$, where $\pi = 3.14159\ldots$ is the number.

(ii) The logistic distribution has mean 0 and variance $\pi^2/3$; so if the latent error of Theorem 7.9 has standard deviation s, a logistic error gives the coefficients $\boldsymbol{\beta}_L = \pi\boldsymbol{\beta}/(\sqrt{3}\, s)$ and a normal one $\boldsymbol{\beta}_P = \boldsymbol{\beta}/s$, and $\boldsymbol{\beta}_L = \pi\boldsymbol{\beta}_P/\sqrt{3}$.

Proof. (i) $\Lambda'(0) = \Lambda(0)\{1 - \Lambda(0)\} = 1/4$ and $\Phi'(0) = 1/\sqrt{2\pi}$, so the slopes are $\beta_L/4$ and $\beta_P/\sqrt{2\pi}$.

(ii) The density $\lambda(z) = \Lambda'(z) = e^{-z}/(1+e^{-z})^2$ is even by Proposition 7.8, and $\lambda(z) \leqslant e^{-|z|}$, so every moment exists and the mean is 0. Integrating by parts, with $1-\Lambda$ an antiderivative of $-\lambda$ and $z^2\{1-\Lambda(z)\} \leqslant z^2 e^{-z} \to 0$, $\int_0^\infty z^2\lambda(z)\,\mathrm{d}z = 2\int_0^\infty z\{1-\Lambda(z)\}\,\mathrm{d}z$. For $z > 0$, $1-\Lambda(z) = e^{-z}/(1+e^{-z}) = \sum_{m\geqslant 1}(-1)^{m+1}e^{-mz}$, a geometric series whose partial sums $e^{-z}\{1-(-e^{-z})^N\}/(1+e^{-z})$ lie

in $[0, e^{-z}]$; with the dominating function ze^{-z}, dominated convergence integrates it term by term:

$$\text{var} = 2\int_0^\infty z^2\lambda(z)\,\mathrm{d}z = 4\sum_{m\geqslant 1}(-1)^{m+1}\int_0^\infty ze^{-mz}\,\mathrm{d}z = 4\sum_{m\geqslant 1}\frac{(-1)^{m+1}}{m^2} = \frac{\pi^2}{3}$$

by the prerequisites' lemma **sum of reciprocal squares**. By Theorem 7.9, a logistic latent error of standard deviation s is $\sigma\varepsilon$ with ε standard logistic and $\sigma = \sqrt{3}\,s/\pi$, which gives $\Lambda(\mathbf{x}'\boldsymbol{\beta}/\sigma)$, and a normal one is sZ, which gives $\Phi(\mathbf{x}'\boldsymbol{\beta}/s)$. □

Example 7.6 (Similarity of the two functions; FREES §11.2.1, Figure 11.1)**.** FREES's comparison of $\Lambda(\pi z/\sqrt{3})$ with $\Phi(z)$, the two factors of Proposition 7.12, and the ratio $\text{logit}(p)/\Phi^{-1}(p)$ over $0.1 \leqslant p \leqslant 0.9$:

```
z <- seq(-4, 4, length = 2001); dl <- plogis(z * pi / sqrt(3)) - pnorm(z)
pp <- seq(0.1, 0.9, length = 801); pp <- pp[abs(pp - 0.5) > 1e-9]; rr <- qlogis(pp)
 ↪  / qnorm(pp)
round(c(max_abs_difference = max(abs(dl)), at_z = abs(z[which.max(abs(dl))]),
```

```
        slope_factor = 4 / sqrt(2 * pi), sd_factor = pi / sqrt(3), ratio_min =
        ↪  min(rr), ratio_max = max(rr)), 4)
```

```
## max_abs_difference              at_z       slope_factor
##             0.0227            0.6840             1.5958
##          sd_factor          ratio_min          ratio_max
##             1.8138             1.5958             1.7145
```

```
op <- par(mfrow = c(1, 2), mar = c(3.2, 3.8, 0.6, 0.6), mgp = c(2.1, 0.6, 0))
plot(z, pnorm(z), type = "l", ylab = "distribution function"); lines(z, plogis(z *
 ↪  pi / sqrt(3)), lty = 2)
plot(z, dnorm(z), type = "l", ylab = "density", ylim = c(0, 0.47))
lines(z, pi / sqrt(3) * dlogis(z * pi / sqrt(3)), lty = 2)
par(op)
```

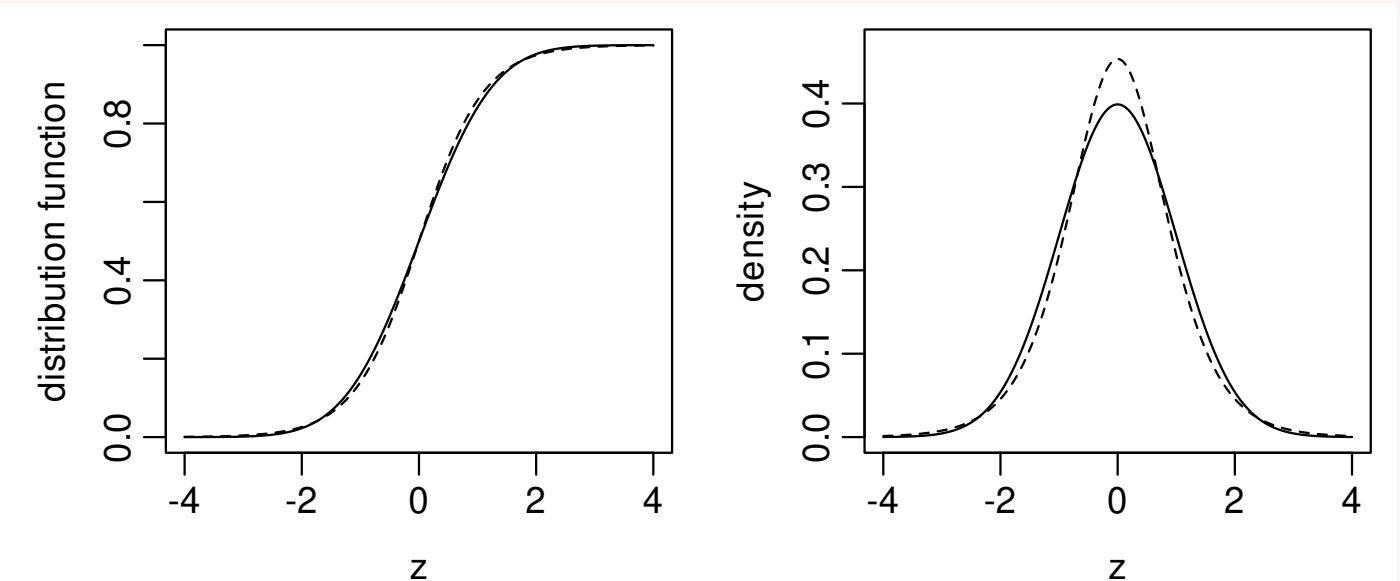


Figure 7.3: The standard normal (solid) and the logistic rescaled to unit variance (dashed): distribution functions (FREES Figure 11.1) and densities.

The rescaled logistic and the normal distribution functions differ by at most 0.0227, at $|z| = 0.684$, and on $0.1 \leqslant p \leqslant 0.9$ the ratio $\text{logit}(p)/\Phi^{-1}(p)$ lies between 1.5958, the slope factor of

Proposition 7.12(i), and 1.7145: over this range the two links are nearly proportional.

3 Inference for Logistic and Probit Regression

FREES §11.3; ISLR §4.3.2–4.3.4 (PA). SRM learning outcomes 2(e), 2(f).

3.1 The likelihood and its maximiser

Definition 7.13 (Log-likelihood; FREES §11.3.1, ISLR (4.5))**.** The log-likelihood of a binary regression is

$$L(\boldsymbol{\beta}) = \sum_{i=1}^{n} \left[y_i \ln \pi(\mathbf{x}_i'\boldsymbol{\beta}) + (1 - y_i) \ln\{1 - \pi(\mathbf{x}_i'\boldsymbol{\beta})\} \right],$$

the logarithm of the probability of the observed responses as a function of $\boldsymbol{\beta}$, and a maximiser $\mathbf{b}$ is a **maximum likelihood estimate**. When $\pi(-z) = 1 - \pi(z)$, as for the logit and the probit, the ith term is $\ln \pi(s_i \mathbf{x}_i'\boldsymbol{\beta})$ with $s_i = 2y_i - 1$.

Theorem 7.14 (Score and information; FREES §11.3.1, §11.3.2)**.** Let π be twice continuously differentiable with $\pi' > 0$, and write $\eta_i = \mathbf{x}_i'\boldsymbol{\beta}$ and $\pi_i = \pi(\eta_i)$. Then

$$\nabla L(\boldsymbol{\beta}) = \sum_{i=1}^{n} \mathbf{x}_i (y_i - \pi_i) \frac{\pi'(\eta_i)}{\pi_i(1-\pi_i)}, \qquad \mathcal{I}(\boldsymbol{\beta}) = \operatorname{var} \nabla L(\boldsymbol{\beta}) = \sum_{i=1}^{n} \frac{\pi'(\eta_i)^2}{\pi_i(1-\pi_i)} \mathbf{x}_i \mathbf{x}_i',$$

and $\mathcal{I}(\boldsymbol{\beta}) = -\mathsf{E}\, \nabla^2 L(\boldsymbol{\beta})$, the **information matrix**. For the logit, $\nabla L(\boldsymbol{\beta}) = \mathbf{X}'(\mathbf{Y} - \boldsymbol{\pi})$ and $-\nabla^2 L(\boldsymbol{\beta}) = \mathbf{X}'\mathbf{W}\mathbf{X} = \mathcal{I}(\boldsymbol{\beta})$ with $\mathbf{W} = \operatorname{diag}\{\pi_i(1-\pi_i)\}$, whatever the responses. For the probit, $-\nabla^2 L(\boldsymbol{\beta}) = \sum_i w_i \mathbf{x}_i \mathbf{x}_i'$ with the **observed weights**

$$w_i = y_i \lambda_1(\eta_i)\{\lambda_1(\eta_i) + \eta_i\} + (1 - y_i)\lambda_0(\eta_i)\{\lambda_0(\eta_i) - \eta_i\}, \qquad \lambda_1 = \frac{\varphi}{\Phi}, \quad \lambda_0 = \frac{\varphi}{1-\Phi}.$$

Proof. As $\partial \eta_i / \partial \boldsymbol{\beta} = \mathbf{x}_i$, the chain rule gives

$$\nabla L = \sum_i \mathbf{x}_i \left\{ \frac{y_i}{\pi_i} - \frac{1-y_i}{1-\pi_i} \right\} \pi'(\eta_i) = \sum_i \mathbf{x}_i \frac{y_i(1-\pi_i) - (1-y_i)\pi_i}{\pi_i(1-\pi_i)} \pi'(\eta_i),$$

with numerator $y_i - \pi_i$. The summands are independent with mean zero, and $\operatorname{var} y_i = \pi_i(1-\pi_i)$ by Proposition 7.2, so the variance of the score is the displayed sum; it equals $-\mathsf{E}\,\nabla^2 L$ by the prerequisites' lemma **score identities**, the responses taking two values. For the logit, $\pi' = \pi(1-\pi)$ by Proposition 7.8, which reduces the score to $\sum_i \mathbf{x}_i(y_i - \pi_i)$, and differentiating again, $\nabla^2 L = -\sum_i \pi_i(1-\pi_i)\mathbf{x}_i\mathbf{x}_i'$, free of the responses. For the probit, with $\varphi'(\eta) = -\eta\varphi(\eta)$, $(\ln\Phi)'' = (\varphi'\Phi - \varphi^2)/\Phi^2 = -\lambda_1(\lambda_1 + \eta)$ and $\{\ln(1-\Phi)\}'' = -\{\varphi'(1-\Phi) + \varphi^2\}/(1-\Phi)^2 = -\lambda_0(\lambda_0 - \eta)$, so the second derivative of the ith term in η_i is $-w_i$. □

Theorem 7.15 (Concavity)**.** For the logit and the probit, each term of L is concave in $\boldsymbol{\beta}$, and L is strictly concave when $\operatorname{rank}(\mathbf{X}) = k+1$.

Proof. Logit. $\ln\Lambda(\eta) = -\ln(1+e^{-\eta})$ has second derivative $-\Lambda(\eta)\{1-\Lambda(\eta)\} < 0$, and $\ln\{1-\Lambda(\eta)\} = \ln\Lambda(-\eta)$ has the same second derivative, the chain rule contributing $(-1)^2$.
Probit. By Unit 5's lemma **truncated normal moments**, λ_0 is differentiable with $\lambda_0' \in (0,1)$; as $\{\ln(1-\Phi)\}' = -\lambda_0$, the function $\ln(1-\Phi)$ has second derivative $-\lambda_0' < 0$, and so does $\ln\Phi(\eta) = \ln\{1-\Phi(-\eta)\}$, at $-\eta$.

Both. So the ith term $\ell_i(\eta)$ has $\ell_i'' < 0$. Along a line, $t \mapsto \ell_i(\mathbf{x}_i'\boldsymbol{\beta} + t\mathbf{x}_i'\mathbf{d})$ has second derivative $\ell_i''(\cdot)(\mathbf{x}_i'\mathbf{d})^2 \leqslant 0$, and $t \mapsto L(\boldsymbol{\beta} + t\mathbf{d})$ has second derivative $-\sum_i c_i(\mathbf{x}_i'\mathbf{d})^2$ with $c_i = -\ell_i''(\mathbf{x}_i'\boldsymbol{\beta} + t\mathbf{x}_i'\mathbf{d}) > 0$, negative iff $\mathbf{X}\mathbf{d} \neq \mathbf{0}$, which holds for every $\mathbf{d} \neq \mathbf{0}$ iff $\operatorname{rank}(\mathbf{X}) = k + 1$. By the prerequisites' lemma **convexity from the derivative**, each term is concave and, when $\operatorname{rank}(\mathbf{X}) = k + 1$, L is strictly concave. □

Corollary 7.16 (The logit score equations; FREES §11.3.1)**.** A logit maximum likelihood estimate $\mathbf{b}$, with fitted values $\hat{y}_i = \Lambda(\mathbf{x}_i'\mathbf{b})$, satisfies $\sum_i x_{ij}\hat{y}_i = \sum_i x_{ij}y_i$ for every j; the intercept gives $\sum_i \hat{y}_i = \sum_i y_i$. If $\mathbf{x}_i = (1, x_i)'$ with $x_i \in \{0, 1\}$, and the n_1 observations with $x_i = 0$ and the n_2 with $x_i = 1$ have means $\bar{y}_1, \bar{y}_2 \in (0, 1)$, then the estimate is $b_0 = \operatorname{logit}\bar{y}_1$, $b_1 = \operatorname{logit}\bar{y}_2 - \operatorname{logit}\bar{y}_1$, and e^{b_1} is the sample odds ratio.

Proof. A maximiser of the differentiable L has $\nabla L = \mathbf{X}'(\mathbf{Y} - \hat{\mathbf{Y}}) = \mathbf{0}$ by Theorem 7.14, which read coordinate by coordinate is the first statement, and the intercept column is $\mathbf{1}$. In the two-group case the fitted values are $\Lambda(b_0)$ and $\Lambda(b_0 + b_1)$ in the two groups, and the score equations for the intercept and for x_i say $n_1\Lambda(b_0) + n_2\Lambda(b_0 + b_1) = n_1\bar{y}_1 + n_2\bar{y}_2$ and $n_2\Lambda(b_0 + b_1) = n_2\bar{y}_2$,

so $\Lambda(b_0) = \bar{y}_1$ and $\Lambda(b_0 + b_1) = \bar{y}_2$, solved by the stated values by Proposition 7.8. These solve $\nabla L = \mathbf{0}$, and a stationary point of the concave L (Theorem 7.15) is a maximiser by the prerequisites' proposition **first-order condition**; $\operatorname{rank}(\mathbf{X}) = 2$ makes L strictly concave, so it is the only one, as two maximisers would give a larger value at their midpoint. Finally $e^{b_1} = \{\bar{y}_2/(1-\bar{y}_2)\}/\{\bar{y}_1/(1-\bar{y}_1)\}$. □

Example 7.7 (Sex alone; FREES §11.3.1)**.** The logit fit of `POSEXP` on `GENDER` for the data of Example 7.1:

```
g1 <- glm(POSEXP ~ GENDER, family = binomial, data = H)
round(c(b0 = coef(g1)[[1]], logit_men = qlogis(p_m), b1 = coef(g1)[[2]], exp_b1 =
  exp(coef(g1)[[2]]),
        sample_odds_ratio = (p_w / (1 - p_w)) / (p_m / (1 - p_m))), 4)
```

```
##              b0       logit_men              b1
##         -3.0204         -3.0204          0.9009
##          exp_b1 sample_odds_ratio
##          2.4617          2.4617
```

$b_0 = -3.0204$ is the logit of the men's proportion, and $e^{b_1} = 2.4617$ is the sample odds ratio of

Example 7.5, as Corollary 7.16 states.

Theorem 7.17 (Existence and separation)**.** Let L be a logit or probit log-likelihood with $\operatorname{rank}(\mathbf{X}) = k+1$, and $s_i = 2y_i - 1$.

(i) If some $\mathbf{c} \neq \mathbf{0}$ has $s_i \mathbf{x}_i' \mathbf{c} \geqslant 0$ for every i, no maximum likelihood estimate exists. If all these inequalities are strict (the data are **separated**), $L(t\mathbf{c})$ increases to $\sup L = 0$ as $t \to \infty$.

(ii) Otherwise L has exactly one maximiser.

Proof. By Definition 7.13, $L(\boldsymbol{\beta}) = \sum_i \ln \pi(s_i \mathbf{x}_i' \boldsymbol{\beta})$ with $\ln \pi$ strictly increasing and negative.
(i) For any $\boldsymbol{\beta}$, $L(\boldsymbol{\beta} + t\mathbf{c}) = \sum_i \ln \pi(s_i \mathbf{x}_i' \boldsymbol{\beta} + t s_i \mathbf{x}_i' \mathbf{c})$ is nondecreasing in t, and strictly increasing because some $s_i \mathbf{x}_i' \mathbf{c} > 0$: otherwise $\mathbf{X}\mathbf{c} = \mathbf{0}$, against $\operatorname{rank}(\mathbf{X}) = k+1$. So no $\boldsymbol{\beta}$ maximises L. With strict inequalities each term of $L(t\mathbf{c})$ increases to $\ln 1 = 0$, while $L < 0$ everywhere.
(ii) Now every unit vector $\mathbf{c}$ has some $s_i \mathbf{x}_i' \mathbf{c} < 0$, so $m(\mathbf{c}) = \max_i(-s_i \mathbf{x}_i' \mathbf{c}) > 0$. The function m is continuous on the unit sphere, which is compact by the prerequisites' theorem **Heine–Borel**, so it attains a minimum $m_0 > 0$ by the prerequisites' theorem **uniform continuity on compacts**. Keeping only the term at which the maximum is attained, and using that $-\ln \pi$ is decreasing and

nonnegative,

$$-L(t\mathbf{c}) = \sum_i \{-\ln \pi(ts_i \mathbf{x}_i' \mathbf{c})\} \geqslant -\ln \pi(-tm(\mathbf{c})) \geqslant -\ln \pi(-tm_0) \longrightarrow \infty$$

as $t \to \infty$, uniformly in $\mathbf{c}$. So $-L$ is continuous and coercive and has a minimiser by the prerequisites' lemma **coercive minimisation**; it is strictly convex by Theorem 7.15, so the minimiser is unique by the same lemma. □

Example 7.8 (A separated sample)**.** Six observations with $y_i = \mathbb{1}\{x_i > 3.5\}$ are separated by $\mathbf{c} = (-3.5, 1)'$:

```
xs <- 1:6; ys <- c(0, 0, 0, 1, 1, 1)
gs <- suppressWarnings(glm(ys ~ xs, family = binomial))
round(c(slope = coef(gs)[[2]], se = coef(summary(gs))[2, 2], iterations = gs$iter,
        max_fitted_y0 = max(fitted(gs)[ys == 0]), min_fitted_y1 = min(fitted(gs)[ys
        ↪  == 1])), 6)
```

```
##         slope            se    iterations max_fitted_y0
##         47.23     115264.41         25.00          0.00
```

```
## min_fitted_y1
##          1.00
```

```
Ls <- function(t) sum(plogis((2 * ys - 1) * t * (xs - 3.5), log.p = TRUE))
round(c(t1 = Ls(1), t5 = Ls(5)), 4); signif(c(t20 = Ls(20)), 3)
```

```
##      t1      t5
## -1.5088 -0.1589
##        t20
## -0.0000908
```

`glm` stops after 25 iterations with slope 47.23, standard error 115264.41 and fitted values 0.00 and 1.00 to the printed digits: a symptom of Theorem 7.17(i), not an estimate. Along **c** the log-likelihood rises towards 0: -1.5088, -0.1589 and -0.0000908 at $t = 1$, 5 and 20.

3.2 Fisher scoring (beyond the syllabus)

Beyond the syllabus: FREES §11.9.2, the chapter's technical supplement, which is outside the reading list.

Theorem 7.18 (Fisher scoring is iteratively reweighted least squares)**.** Let π be as in Theorem 7.14, $\operatorname{rank}(\mathbf{X}) = k+1$, and at a current value $\boldsymbol{\beta}^{(m)}$ let $\eta_i = \mathbf{x}_i'\boldsymbol{\beta}^{(m)}$, $\pi_i = \pi(\eta_i)$, $f_i = \pi'(\eta_i)$, $\omega_i = f_i^2/\{\pi_i(1-\pi_i)\}$ and $z_i = \eta_i + (y_i - \pi_i)/f_i$. The Fisher scoring step $\boldsymbol{\beta}^{(m)} + \mathcal{I}(\boldsymbol{\beta}^{(m)})^{-1}\nabla L(\boldsymbol{\beta}^{(m)})$ is the weighted least squares fit $(\mathbf{X}'\boldsymbol{\Omega}\mathbf{X})^{-1}\mathbf{X}'\boldsymbol{\Omega}\mathbf{z}$, $\boldsymbol{\Omega} = \operatorname{diag}(\omega_i)$. For the logit it is also the Newton–Raphson step $\boldsymbol{\beta}^{(m)} - \{\nabla^2 L(\boldsymbol{\beta}^{(m)})\}^{-1}\nabla L(\boldsymbol{\beta}^{(m)})$.

Proof. By Theorem 7.14, $\mathcal{I} = \mathbf{X}'\boldsymbol{\Omega}\mathbf{X}$, invertible as $\operatorname{rank}(\mathbf{X}) = k+1$ and every $\omega_i > 0$, and $\nabla L = \sum_i \mathbf{x}_i \omega_i (y_i - \pi_i)/f_i = \mathbf{X}'\boldsymbol{\Omega}(\mathbf{z} - \mathbf{X}\boldsymbol{\beta}^{(m)})$. So $\boldsymbol{\beta}^{(m)} + (\mathbf{X}'\boldsymbol{\Omega}\mathbf{X})^{-1}\mathbf{X}'\boldsymbol{\Omega}(\mathbf{z} - \mathbf{X}\boldsymbol{\beta}^{(m)}) = (\mathbf{X}'\boldsymbol{\Omega}\mathbf{X})^{-1}\mathbf{X}'\boldsymbol{\Omega}\mathbf{z}$. For the logit $-\nabla^2 L = \mathcal{I}$ by the same theorem. □

3.3 Large-sample inference

Theorem 7.19 (Large-sample distribution; FREES §11.3.2)**.** Let the responses follow the logit or the probit model with parameter $\boldsymbol{\beta}_0$, and let the design satisfy
(D1) $|\mathbf{x}_i| \leqslant M$ for every i, with M free of n;
(D2) $n^{-1}\mathcal{I}(\boldsymbol{\beta}_0) \to \mathbf{F}$, positive definite.
Then with probability tending to one the maximum likelihood estimate $\mathbf{b}$ exists and is unique, $\mathbf{b} \xrightarrow{p} \boldsymbol{\beta}_0$, $\sqrt{n}(\mathbf{b}-\boldsymbol{\beta}_0) \xrightarrow{d} N(\mathbf{0}, \mathbf{F}^{-1})$, and both $n^{-1}\mathcal{I}(\mathbf{b})$ and $-n^{-1}\nabla^2 L(\mathbf{b})$ converge in probability to $\mathbf{F}$.

Proof. The hypotheses of the prerequisites' theorem **concave maximum likelihood** are checked with $\Theta = \mathbb{R}^{k+1}$ and $\ell_i(\boldsymbol{\beta}) = y_i \ln \pi(\mathbf{x}_i'\boldsymbol{\beta}) + (1-y_i)\ln\{1-\pi(\mathbf{x}_i'\boldsymbol{\beta})\}$. Write $c(\eta; y)$ for minus the second derivative in η of $y\ln\pi(\eta) + (1-y)\ln\{1-\pi(\eta)\}$: $\Lambda(\eta)\{1-\Lambda(\eta)\}$ for the logit and the observed weight of Theorem 7.14 for the probit. By the prerequisites' theorem **Cauchy–Schwarz**, $|\mathbf{x}_i'\boldsymbol{\beta}| \leqslant M(|\boldsymbol{\beta}_0|+1)$ whenever $|\boldsymbol{\beta}-\boldsymbol{\beta}_0| \leqslant 1$; call this interval of η values J.
Concavity and smoothness. Each ℓ_i is concave by Theorem 7.15 and twice continuously differentiable.

The score. By Theorem 7.14, $\nabla \ell_i(\boldsymbol{\beta}_0) = \mathbf{x}_i r(\eta_i)(y_i - \pi_i)$ with $r = \pi'/\{\pi(1-\pi)\}$, which has mean zero and variance $\pi'(\eta_i)^2 \mathbf{x}_i \mathbf{x}_i'/\{\pi_i(1-\pi_i)\}$, so the average variance is $n^{-1}\mathcal{I}(\boldsymbol{\beta}_0) \to \mathbf{F}$ by (D2). The continuous r is bounded on the compact J, by R_0 say, by the prerequisites' theorem **uniform continuity on compacts**, so $|\nabla \ell_i(\boldsymbol{\beta}_0)| \leqslant MR_0$, and the prerequisites' corollary **bounded summands with unequal variances** gives $n^{-1/2}\nabla L(\boldsymbol{\beta}_0) \xrightarrow{d} N(\mathbf{0}, \mathbf{F})$.

The curvature. $-\nabla^2 L(\boldsymbol{\beta}_0) = \sum_i c(\eta_i; y_i)\mathbf{x}_i\mathbf{x}_i'$. For the logit it equals $\mathcal{I}(\boldsymbol{\beta}_0)$. For the probit, $\mathsf{E}\, c(\eta_i; y_i) = \pi_i\lambda_1(\lambda_1 + \eta_i) + (1-\pi_i)\lambda_0(\lambda_0 - \eta_i) = \varphi^2/\Phi + \varphi^2/(1-\Phi)$ at η_i, which is the weight of $\mathcal{I}(\boldsymbol{\beta}_0)$; each entry of $n^{-1}\{-\nabla^2 L(\boldsymbol{\beta}_0) - \mathcal{I}(\boldsymbol{\beta}_0)\}$ is an average of independent mean-zero terms bounded by a constant times M^2, so its variance is of order $1/n$ and it tends to 0 in probability by the prerequisites' theorem **Markov, Chebyshev**. In both cases $-n^{-1}\nabla^2 L(\boldsymbol{\beta}_0) \xrightarrow{p} \mathbf{F}$. The derivative of $c(\cdot; y)$ is continuous, hence bounded on J by some K_0 for $y = 0, 1$ by the same theorem, so the prerequisites' mean value theorem gives $|c(\mathbf{x}_i'\boldsymbol{\beta}; y_i) - c(\mathbf{x}_i'\boldsymbol{\beta}_0; y_i)| \leqslant K_0 M |\boldsymbol{\beta} - \boldsymbol{\beta}_0|$ and, with $\|\mathbf{x}_i\mathbf{x}_i'\| = |\mathbf{x}_i|^2 \leqslant M^2$,

$$\|\nabla^2 L(\boldsymbol{\beta}) - \nabla^2 L(\boldsymbol{\beta}_0)\| \leqslant nK_0M^3|\boldsymbol{\beta} - \boldsymbol{\beta}_0|, \qquad |\boldsymbol{\beta} - \boldsymbol{\beta}_0| \leqslant 1.$$

The same argument applies to $\mathcal{I}(\boldsymbol{\beta})$, whose weights $\pi'^2/\{\pi(1-\pi)\}$ have a continuous derivative.

Conclusion. The limiting curvature equals the limiting variance of the score, so the theorem gives existence with probability tending to one, consistency, the normal limit with variance $\mathbf{F}^{-1}$, and $-n^{-1}\nabla^2 L(\mathbf{b}) \xrightarrow{p} \mathbf{F}$; then $\|n^{-1}\mathcal{I}(\mathbf{b}) - n^{-1}\mathcal{I}(\boldsymbol{\beta}_0)\| \leqslant K_0' M^3 |\mathbf{b} - \boldsymbol{\beta}_0| \xrightarrow{p} 0$ for the constant K_0' of $\mathcal{I}$. For uniqueness, (D2) makes $n^{-1}\mathbf{d}'\mathcal{I}(\boldsymbol{\beta}_0)\mathbf{d} > 0$ for every $\mathbf{d} \neq \mathbf{0}$ once n is large, which forces $\operatorname{rank}(\mathbf{X}) = k+1$; L is then strictly concave by Theorem 7.15, and two maximisers would give a larger value at their midpoint. □

Remark (FREES §11.3.2)**.** So $\mathbf{b}$ is approximately $N(\boldsymbol{\beta}_0, \mathcal{I}(\mathbf{b})^{-1})$, and $\operatorname{se}(b_j)$ is the square root of the $(j+1)$st diagonal element of $\mathcal{I}(\mathbf{b})^{-1}$; FREES's text says "of this matrix", and its errata correct this to the inverse. The ratio $z = b_j / \operatorname{se}(b_j)$ is referred to $N(0,1)$ when $\beta_j = 0$; R's `glm` computes se from the expected information $\mathcal{I}(\mathbf{b})$, which for the logit is also the observed $-\nabla^2 L(\mathbf{b})$ but not for the probit, where either may be used.

Definition 7.20 (Wald and likelihood ratio tests; FREES §11.3.2)**.** For H_0: $\mathbf{R}\boldsymbol{\beta} = \mathbf{r}$ with $\mathbf{R}$ a $q \times (k+1)$ matrix of rank q, let $\mathbf{b}_R$ maximise L subject to $\mathbf{R}\boldsymbol{\beta} = \mathbf{r}$. The **Wald** and **likelihood**

ratio statistics are

$$W = (\mathbf{Rb} - \mathbf{r})'\{\mathbf{R}\mathcal{I}(\mathbf{b})^{-1}\mathbf{R}'\}^{-1}(\mathbf{Rb} - \mathbf{r}), \qquad \text{LRT} = 2\{L(\mathbf{b}) - L(\mathbf{b}_R)\}.$$

For the overall hypothesis $\beta_1 = \cdots = \beta_k = 0$, $L(\mathbf{b}_R)$ is the maximised log-likelihood L_0 of the intercept-only model and $q = k$.

Corollary 7.21 (Chi-square limits; FREES §11.3.2)**.** Under Theorem 7.19 and H_0, W and LRT converge in distribution to χ^2_q. For $H_0\colon \beta_j = 0$, $W = z^2$.

Proof. By Theorem 7.19 and the prerequisites' theorems **continuous mapping** and **closure under affine maps**, $\sqrt{n}(\mathbf{Rb} - \mathbf{r}) = \mathbf{R}\sqrt{n}(\mathbf{b} - \boldsymbol{\beta}_0) \xrightarrow{d} N(\mathbf{0}, \mathbf{RF}^{-1}\mathbf{R}')$, with $\mathbf{RF}^{-1}\mathbf{R}'$ positive definite as $\mathbf{R}$ has rank q; and $\mathbf{R}\{n^{-1}\mathcal{I}(\mathbf{b})\}^{-1}\mathbf{R}' \xrightarrow{p} \mathbf{RF}^{-1}\mathbf{R}'$ by the prerequisites' lemma **inverses of convergent positive definite matrices** and theorem **continuous mapping**. As $W = \{\sqrt{n}(\mathbf{Rb}-\mathbf{r})\}'[\mathbf{R}\{n^{-1}\mathcal{I}(\mathbf{b})\}^{-1}\mathbf{R}']^{-1}\{\sqrt{n}(\mathbf{Rb}-\mathbf{r})\}$, the prerequisites' theorem **quadratic forms of asymptotically normal vectors** gives $W \xrightarrow{d} \chi^2_q$. For the likelihood ratio, complete $\mathbf{R}$ to an invertible matrix $\mathbf{T}$ whose last q rows are $\mathbf{R}$ and write $\boldsymbol{\gamma} = \mathbf{T}\boldsymbol{\beta}$: the model is a binary regression

in $\boldsymbol{\gamma}$ with rows $(\mathbf{T}')^{-1}\mathbf{x}_i$, which satisfy (D1) and (D2), and H_0 fixes the last q coordinates of $\boldsymbol{\gamma}$. The restricted problem is a binary regression in the first $k+1-q$ coordinates with a known offset, and the proof of Theorem 7.19 applies to it unchanged. The prerequisites' theorem **likelihood ratio** then gives LRT $\xrightarrow{d} \chi^2_q$. For $q=1$ and $\mathbf{R}$ the jth unit row, $W = b_j^2/[\mathcal{I}(\mathbf{b})^{-1}]_{jj} = z^2$. □

Proposition 7.22 (The likelihood ratio in the normal linear model)**.** Under A1–A5 and the true hypothesis that q given slopes are zero, with k fixed, the likelihood ratio statistic of the submodel without them against the full model, twice the difference of the maximised normal log-likelihoods, converges in distribution to χ^2_q as $n \to \infty$.

Proof. With $m = n-k-1$, Unit 5's proposition **likelihood ratio and** F gives $\mathrm{LR} = n\ln(1+qF/m)$ with $F = \{(\mathrm{RSS}_R - \mathrm{RSS}_U)/q\}/(\mathrm{RSS}_U/m)$, which by Unit 3's corollary **extra sum of squares** is the statistic of Unit 3's theorem **the general linear hypothesis**: defined with probability one, $\mathrm{RSS}_U > 0$, and under H_0 of law $F_{q,m}$, the law of $(A/q)/(B/m)$ with $A \sim \chi^2_q$ and $B \sim \chi^2_m$ independent. By the prerequisites' proposition **moments of** χ^2, B/m has mean 1 and variance $2/m$, so $B/m \xrightarrow{p} 1$ by the prerequisites' theorem **Markov, Chebyshev** and

$qF \xrightarrow{d} \chi^2_q$ by the prerequisites' theorem **Slutsky**; hence $u_n = qF/m \xrightarrow{d} 0$, so $u_n \xrightarrow{p} 0$ by the prerequisites' proposition **convergence to a constant**. With $h(u) = \ln(1+u)/u$ for $u > -1$, $u \neq 0$, and $h(0) = 1$, continuous, $\mathrm{LR} = (n/m)\, qF\, h(u_n)$, and the same theorem with $n/m \to 1$ and $h(u_n) \xrightarrow{p} 1$, by the prerequisites' theorem **continuous mapping**, gives $\mathrm{LR} \xrightarrow{d} \chi^2_q$. □

Definition 7.23 (Information criterion; FREES §11.4, Table 11.5)**.** For a model fitted by maximum likelihood with d estimated parameters, $\mathrm{AIC} = -2L(\mathbf{b}) + 2d$. For ordinary linear models the SRM syllabus uses instead the form of Unit 5's definition C_p**, AIC, BIC and adjusted** R^2.

Definition 7.24 (Measures of fit; FREES §11.3.2)**.** With L_0 the maximised intercept-only log-likelihood, the **max-scaled** R^2 is $R^2_{ms} = R^2/R^2_{\max}$, where $R^2 = 1 - \exp\{2(L_0 - L(\mathbf{b}))/n\}$ and $R^2_{\max} = 1 - \exp(2L_0/n)$; FREES prints R^2 without the factor 2 in the exponent, and its errata restore it. The **pseudo-**R^2 is $\{L(\mathbf{b}) - L_0\}/(L_{\max} - L_0)$, with $L_{\max}$ the supremum of the log-likelihood over all fitted probabilities in $(0,1)^n$.

Proposition 7.25 (Range of the measures; FREES §11.3.2)**.** For binary responses, not all equal, a model with an intercept and its maximum likelihood estimate $\mathbf{b}$, $L_{\max} = 0$ and $L_0 \leqslant L(\mathbf{b}) < 0$, so the pseudo-$R^2$ and R^2_{ms} lie in $[0, 1]$, and LRT $= 2\{L(\mathbf{b}) - L_0\} = -2L_0 \times$ pseudo-R^2.

Proof. Each term $y_i \ln p_i + (1 - y_i) \ln(1 - p_i)$ is negative for $p_i \in (0, 1)$ and tends to 0 as $p_i \to y_i$, so $L_{\max} = 0$ and $L(\mathbf{b}) < 0$. The intercept-only model is the model with $\beta_1 = \cdots = \beta_k = 0$, so $L(\mathbf{b}) \geqslant L_0$. Hence $\{L(\mathbf{b}) - L_0\}/(0 - L_0) \in [0, 1]$, and $2(L_0 - L(\mathbf{b}))/n \in [2L_0/n, 0]$ gives $0 \leqslant R^2 \leqslant R^2_{\max}$, with $R^2_{\max} > 0$ as $L_0 < 0$. The last identity is the definition of the pseudo-R^2 with $L_{\max} = 0$. □

Example 7.9 (Job security; FREES §11.3.2, Table 11.3)**.** Valletta's probit of dismissal on $n = 24{,}168$ PSID observations reports, among other coefficients, tenure -0.084, tenure $\times$ time trend 0.003, change in logarithmic sector employment 0.094, and tenure $\times$ that change -0.020, with $-2L(\mathbf{b}) = 7027.8$ and pseudo-R^2 0.097. The tenure effect on the index in year t of the trend, the sector employment effect at the mean tenure 7.8, and the likelihood ratio statistic of Proposition 7.25:

```
L_job <- -7027.8 / 2; L0_job <- L_job / (1 - 0.097)
round(c(tenure_effect_t0 = -0.084, tenure_effect_t16 = -0.084 + 0.003 * 16,
        sector_effect_tenure0 = 0.094, sector_effect_tenure7.8 = 0.094 - 0.020 *
        ↪  7.8,
        L0 = L0_job, LRT = -2 * L0_job * 0.097), 3)
```

```
##       tenure_effect_t0      tenure_effect_t16
##                 -0.084                 -0.036
##  sector_effect_tenure0 sector_effect_tenure7.8
##                  0.094                 -0.062
##                     L0                    LRT
##              -3891.362                754.924
```

The tenure effect on the dismissal index, -0.084 at the start of the trend, is -0.036 in 1992 ($t = 16$): dismissal falls with tenure less steeply than before. The sector employment effect, 0.094 at tenure 0, is -0.062 at the mean tenure 7.8: for experienced men a fall in sector employment raises the dismissal index, FREES's reading. With $L_0 = -3891.362$ the overall likelihood ratio statistic is 754.924.

Example 7.10 (Credit card default; ISLR §4.3.2–4.3.3, Tables 4.1–4.2)**.** The logits of **default** on **balance** and on **student** alone, and ISLR's predictions, from the fits and from ISLR's rounded coefficients:

```
gD2 <- glm(default ~ student, family = binomial, data = Df)
round(coef(summary(lgD)), 5); round(coef(summary(gD2)), 4)
```

```
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept) -10.6513    0.36116  -29.49        0
## balance       0.0055    0.00022   24.95        0
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept)  -3.5041     0.0707  -49.55   0.0000
## studentYes    0.4049     0.1150    3.52   0.0004
```

```
round(c(balance_1000 = predict(lgD, data.frame(balance = 1000), type = "response"),
        balance_2000 = predict(lgD, data.frame(balance = 2000), type = "response"),
        rounded_1000 = plogis(-10.6513 + 0.0055 * 1000),
        rounded_2000 = plogis(-10.6513 + 0.0055 * 2000),
        mean_fitted_logit = mean(fitted(lgD))), 5)
```

```
##    balance_1000.1    balance_2000.1      rounded_1000
##           0.00575           0.58577           0.00576
```

```
##      rounded_2000 mean_fitted_logit
##           0.58630           0.03330
```

```
round(predict(gD2, data.frame(student = c("Yes", "No")), type = "response"), 4)
```

```
##      1      2
## 0.0431 0.0292
```

ISLR's Tables 4.1 and 4.2 are reproduced. A unit of **balance** adds 0.0055 to the log odds of default (Theorem 7.11); its $z = 24.95$ is the Wald ratio of Corollary 7.21. The predicted probability at a balance of 1000 is 0.00575; ISLR's 0.00576 uses the rounded slope 0.0055, and at 2000 the two are 0.5858 and 0.5863, ISLR's 0.586. The logit's fitted probabilities average 0.0333, the default rate, by Corollary 7.16; and with the one indicator **student** the fitted probabilities 0.0431 and 0.0292 are the two groups' default rates, by the same corollary.

Example 7.11 (Confounding; ISLR §4.3.4, Table 4.3, Figure 4.3)**.** With **income** in thousands:

```
Df$income1000 <- Df$income / 1000
gD3 <- glm(default ~ balance + income1000 + student, family = binomial, data = Df)
round(coef(summary(gD3)), 4)
```

```
##              Estimate Std. Error  z value Pr(>|z|)
## (Intercept) -10.8690     0.4923 -22.0801   0.0000
## balance       0.0057     0.0002  24.7376   0.0000
## income1000    0.0030     0.0082   0.3698   0.7115
## studentYes   -0.6468     0.2363  -2.7376   0.0062
```

```
round(predict(gD3, data.frame(balance = 1500, income1000 = 40, student = c("Yes",
    "No")),
                type = "response"), 4)
```

```
##      1      2
## 0.0579 0.1050
```

```
round(rbind(default_rate = tapply(Df$y, Df$student, mean),
            mean_balance = tapply(Df$balance, Df$student, mean)), 4)
```

```
##                    No      Yes
## default_rate   0.0292   0.0431
## mean_balance 771.7704 987.8182
```

```
bins <- cut(Df$balance, c(0, 500, 1000, 1500, 2000, 2700), include.lowest = TRUE)
round(tapply(Df$y, list(bins, Df$student), mean), 4)
```

```
##                       No    Yes
## [0,500]           0.0000 0.0000
## (500,1e+03]       0.0022 0.0000
## (1e+03,1.5e+03] 0.0291 0.0190
## (1.5e+03,2e+03] 0.2568 0.1745
## (2e+03,2.7e+03] 0.8372 0.7143
```

ISLR's Table 4.3 is reproduced. The `student` coefficient is $+0.405$ alone and -0.647 given `balance` and `income`: students default more often overall, 4.31% against 2.92%, but at the same balance no more often: in every balance band of the table their default rate is at or below the non-students', because students carry larger balances, 988 against 772 on average. This is the same mechanism as Unit 3's theorem **omitted variables**, whose bias formula is for least squares and does not carry over to the logit. ISLR's predictions 0.058 and 0.105 for a student and a non-student with balance 1500 and income 40 are 0.0579 and 0.1050 from the fit.

4 Application: Medical Expenditures

FREES §11.4. SRM learning outcomes 2(c), 2(e).

Example 7.12 (Positive expenditure by explanatory variable; FREES §11.4, Table 11.4)**.** For each level of each explanatory variable, the percentage of the $n = 2000$ persons at that level and the percentage of them with positive inpatient expenditure, rounded half up as in FREES:

```
pct1 <- function(k, m) floor(1000 * k / m + 0.5) / 10
vars <- c("GENDER", "RACE", "REGION", "EDUC", "PHSTAT", "MNHPOOR", "ANYLIMIT",
 ↪  "INCOME", "insure")
t114 <- do.call(rbind, lapply(vars, function(v) { k <- table(H[[v]]); s <-
 ↪  tapply(H$POSEXP, H[[v]], sum)
  data.frame(variable = v, level = names(k), pct_data = pct1(as.numeric(k),
   ↪  nrow(H)),
             pct_positive = pct1(as.numeric(s), as.numeric(k))) }))
print(t114, row.names = FALSE)
```

```
##  variable     level pct_data pct_positive
##    GENDER         0     47.3          4.7
```

```
##    GENDER         1     52.7         10.7
##      RACE     WHITE     79.9          7.5
##      RACE     ASIAN      4.3          4.7
##      RACE     BLACK     14.8         10.5
##      RACE     NATIV      1.1         13.6
##    REGION      WEST     27.9          5.4
##    REGION   MIDWEST     19.7          8.7
##    REGION NORTHEAST     14.3         10.1
##    REGION     SOUTH     38.2          8.4
##      EDUC   LHIGHSC     29.5          8.8
##      EDUC   COLLEGE     27.2          6.8
##      EDUC   HIGHSCH     43.3          7.9
##    PHSTAT      EXCE     25.4          5.1
##    PHSTAT      FAIR      9.9          8.1
##    PHSTAT      GOOD     29.9          8.2
##    PHSTAT      POOR      3.8         36.0
##    PHSTAT      VGOO     31.1          6.3
##   MNHPOOR         0     92.6          7.1
##   MNHPOOR         1      7.5         16.8
##  ANYLIMIT         0     77.7          5.9
```

```
##  ANYLIMIT         1     22.3         14.6
##    INCOME      POOR     17.0         13.0
##    INCOME   HINCOME     31.6          5.4
##    INCOME   LINCOME     15.8          8.3
##    INCOME   MINCOME     29.9          7.0
##    INCOME     NPOOR      5.8          9.5
##    insure         0     22.3          3.1
##    insure         1     77.8          9.2
```

```
c(mean_age = round(mean(H$AGE), 1), pct_positive_all = pct1(sum(H$POSEXP), nrow(H)))
```

```
##         mean_age pct_positive_all
##             39.0              7.9
```

The table reproduces FREES's Table 11.4 to the printed digit, the mean age 39.0 and the 7.9% positive overall included. The largest percentages positive are for poor physical health, 36.0%, poor or fair mental health, 16.8%, and an activity limitation, 14.6%; the insured have 9.2% against 3.1% for the uninsured.

Example 7.13 (Logit and probit fits; FREES §11.4, Table 11.5)**.** The full logit model, the reduced logit model without `AGE` and `MNHPOOR`, and the reduced probit model, with z ratios; for

the probit both from the expected information, as **glm** reports, and from the observed information of Theorem 7.14:

```
f_full <- POSEXP ~ AGE + GENDER + RACE + REGION + EDUC + PHSTAT + MNHPOOR + ANYLIMIT + INCOME + insure
full <- glm(f_full, family = binomial, data = H); red <- glm(f_red, family = binomial, data = H)
pro <- glm(f_red, family = binomial(link = "probit"), data = H)
Xp <- model.matrix(pro); ep <- drop(Xp %*% coef(pro)); yp <- H$POSEXP
l1 <- dnorm(ep) / pnorm(ep); l0 <- dnorm(ep) / pnorm(-ep)
z_obs <- coef(pro) / sqrt(diag(solve(crossprod(Xp, (yp * l1 * (l1 + ep) + (1 - yp) * l0 * (l0 - ep)) * Xp))))
nm <- names(coef(full)); pad <- function(v) setNames(v[nm], nm)
round(cbind(full = coef(full), z_full = coef(summary(full))[, 3], reduced = pad(coef(red)),
            z_red = pad(coef(summary(red))[, 3]), probit = pad(coef(pro)),
            z_expected = pad(coef(summary(pro))[, 3]), z_observed = pad(z_obs)), 3)
```

```
##                full z_full reduced   z_red probit
## (Intercept)  -4.239 -8.982  -4.278 -10.094 -2.281
## AGE          -0.001 -0.180      NA      NA     NA
```

```
## GENDER           0.733  3.812   0.732   3.806  0.395
## RACEASIAN       -0.219 -0.411  -0.219  -0.412 -0.108
## RACEBLACK       -0.001 -0.003   0.004   0.019  0.009
## RACENATIV        0.610  0.926   0.612   0.930  0.285
## REGIONMIDWEST    0.524  1.904   0.517   1.883  0.237
## REGIONNORTHEAST  0.609  2.112   0.604   2.098  0.281
## REGIONSOUTH      0.339  1.376   0.328   1.342  0.130
## EDUCCOLLEGE      0.068  0.255   0.070   0.263  0.049
## EDUCHIGHSCH      0.004  0.017   0.009   0.041  0.003
## PHSTATFAIR       0.136  0.375   0.109   0.306  0.079
## PHSTATGOOD       0.376  1.429   0.368   1.405  0.182
## PHSTATPOOR       1.712  4.385   1.652   4.575  0.939
## PHSTATVGOO       0.178  0.667   0.174   0.655  0.094
## MNHPOOR         -0.113 -0.369      NA      NA     NA
## ANYLIMIT         0.564  2.680   0.545   2.704  0.311
## INCOMEHINCOME   -0.921 -3.101  -0.919  -3.162 -0.470
## INCOMELINCOME   -0.411 -1.453  -0.408  -1.449 -0.241
## INCOMEMINCOME   -0.609 -2.315  -0.604  -2.317 -0.314
## INCOMENPOOR     -0.201 -0.528  -0.204  -0.534 -0.146
## insure           1.234  4.047   1.227   4.031  0.579
```

```
##                 z_expected z_observed
## (Intercept)         -11.358    -11.432
## AGE                     NA         NA
## GENDER                4.193      4.178
## RACEASIAN            -0.429     -0.427
## RACEBLACK             0.073      0.073
## RACENATIV             0.785      0.780
## REGIONMIDWEST         1.749      1.754
## REGIONNORTHEAST       1.946      1.950
## REGIONSOUTH           1.080      1.085
## EDUCCOLLEGE           0.361      0.362
## EDUCHIGHSCH           0.030      0.030
## PHSTATFAIR            0.450      0.450
## PHSTATGOOD            1.416      1.412
## PHSTATPOOR            4.740      4.805
## PHSTATVGOO            0.730      0.728
## MNHPOOR                  NA         NA
## ANYLIMIT              3.006      3.022
## INCOMEHINCOME        -3.186     -3.224
## INCOMELINCOME        -1.634     -1.633
```

```
## INCOMEMINCOME       -2.325     -2.345
## INCOMENPOOR         -0.726     -0.721
## insure               4.112      4.147
```

```
print(rbind(log_likelihood = round(sapply(list(full = full, reduced = red, probit = pro), logLik), 2),
            AIC = round(sapply(list(full = full, reduced = red, probit = pro), AIC), 2)), digits = 7)
```

```
##                    full reduced  probit
## log_likelihood -488.69 -488.78 -486.98
## AIC            1021.39 1017.55 1013.97
```

```
LR <- 2 * as.numeric(logLik(full) - logLik(red))
round(c(LRT = LR, p_value = pchisq(LR, 2, lower.tail = FALSE), LRT_rounded_loglik = 2 * (-488.69 + 488.78),
        FREES_p_for_0.36 = pchisq(0.36, 2, lower.tail = FALSE)), 3)
```

```
##               LRT           p_value LRT_rounded_loglik
##             0.166             0.920             0.180
##  FREES_p_for_0.36
##             0.835
```

The three fits reproduce FREES's Table 11.5: the logit coefficients and z ratios to the printed digit and the log-likelihoods -488.69, -488.78 and -486.98. The AIC values 1021.39, 1017.55 and 1013.97 differ in the last digit from FREES's 1021.38, 1017.56 and 1013.96, which are -2 times the rounded log-likelihoods plus $2d$. FREES's probit t ratios are those of the observed information (4.178 for `GENDER`, 4.805 for `PHSTATPOOR`), not of the expected information that `glm` uses (4.193, 4.740). The likelihood ratio statistic for `AGE` and `MNHPOOR` is 0.166, with p-value 0.920 on 2 degrees of freedom; from the rounded log-likelihoods it is 0.180, and FREES's 0.36, with p-value 0.835, is twice that. The conclusion stands: the two variables add nothing.

Example 7.14 (Reading the fits; FREES §11.2.4, §11.3.2, §11.4)**.** The adjusted odds ratio for sex against the raw one of Example 7.5, the ratios of the logit to the probit coefficients that are at least twice their standard errors, the overall likelihood ratio test, and the measures of fit of Definition 7.24:

```
L_red <- as.numeric(logLik(red)); L0_meps <- as.numeric(logLik(glm(POSEXP ~ 1,
 ↪  family = binomial, data = H)))
big <- names(which(abs(coef(summary(red))[, 3]) > 2))
```

```
round(c(odds_ratio_sex_full = exp(coef(full)[["GENDER"]]), exp_0.733 = exp(0.733),
  odds_ratio_sex_raw = (p_w / (1 - p_w)) / (p_m / (1 - p_m))), 3)
```

```
## odds_ratio_sex_full           exp_0.733  odds_ratio_sex_raw
##               2.082               2.081               2.462
```

```
round(coef(red)[big] / coef(pro)[big], 3)
```

```
##     (Intercept)          GENDER REGIONNORTHEAST
##           1.876           1.853           2.149
##      PHSTATPOOR        ANYLIMIT   INCOMEHINCOME
##           1.759           1.751           1.953
##   INCOMEMINCOME          insure
##           1.920           2.120
```

```
R2cs <- 1 - exp(2 * (L0_meps - L_red) / nrow(H)); R2max <- 1 - exp(2 * L0_meps /
  nrow(H))
round(c(L0 = L0_meps, LRT_overall = 2 * (L_red - L0_meps), df = length(coef(red)) -
  1,
        max_scaled_R2 = R2cs / R2max, pseudo_R2 = (L_red - L0_meps) / (0 -
          L0_meps)), 4)
```

```
##            L0   LRT_overall            df max_scaled_R2
##     -550.1810      122.8076       19.0000        0.1407
##     pseudo_R2
##        0.1116
```

```
signif(c(p_value_overall = pchisq(2 * (L_red - L0_meps), length(coef(red)) - 1,
  lower.tail = FALSE)), 3)
```

```
## p_value_overall
##         3.3e-17
```

Adjusting for the other variables, women's odds of hospitalisation are 2.082 times men's, against the raw 2.462; FREES's 2.081 is $e^{0.733}$, from the rounded coefficient. The ratios of the logit to the probit coefficients with $|z| > 2$ lie between 1.751 and 2.149, above the slope factor 1.5958 of Proposition 7.12(i), and six of the eight exceed the standard-deviation factor 1.8138 of (ii). The overall likelihood ratio statistic is 122.8076 on 19 degrees of freedom, with a negligible p-value, and the max-scaled R^2 and the pseudo-R^2 are 0.1407 and 0.1116.

5 Nominal Dependent Variables

FREES §11.5; ISLR §4.3.5 (PA). SRM learning outcomes 2(a), 2(c).

5.1 The generalised logit

Definition 7.26 (Multicategory response; FREES §11.5)**.** A **nominal** response y_i takes one of the unordered values $1, \ldots, c$, $c \geqslant 2$, with $\pi_{ij} = \mathsf{P}(y_i = j)$ and $\sum_j \pi_{ij} = 1$. With $y_{ij} = \mathbb{1}\{y_i = j\}$ the probability of the ith response is $\prod_j \pi_{ij}^{y_{ij}}$, and the log-likelihood of independent responses is $L = \sum_{i=1}^{n} \sum_{j=1}^{c} y_{ij} \ln \pi_{ij}$.

Definition 7.27 (Generalised logit; FREES §11.5.1, ISLR (4.13) (softmax coding))**.** The **generalised logit** model has $V_{ij} = \mathbf{x}_i' \boldsymbol{\beta}_j$ with a parameter vector $\boldsymbol{\beta}_j$ for each category and

$$\pi_{ij} = \frac{\exp(V_{ij})}{\sum_{h=1}^{c} \exp(V_{ih})}, \qquad j = 1, \ldots, c.$$

Proposition 7.28 (Normalisation; FREES §11.5.1, ISLR (4.10)–(4.12), (4.14))**.** Adding the same vector $\mathbf{a}$ to every $\boldsymbol{\beta}_j$ leaves every π_{ij} unchanged, so the normalisation $\boldsymbol{\beta}_c = \mathbf{0}$ loses nothing. Under it, $\ln(\pi_{ij}/\pi_{ic}) = \mathbf{x}_i'\boldsymbol{\beta}_j$, the log odds of category j against the baseline c; if $\operatorname{rank}(\mathbf{X}) = k+1$ the probabilities determine $\boldsymbol{\beta}_1, \ldots, \boldsymbol{\beta}_{c-1}$; and for $c = 2$, $\pi_{i1} = \Lambda(\mathbf{x}_i'\boldsymbol{\beta}_1)$, the logit model.

Proof. Multiplying numerator and denominator by $\exp(\mathbf{x}_i'\mathbf{a})$ changes every $\boldsymbol{\beta}_h$ into $\boldsymbol{\beta}_h + \mathbf{a}$, and $\mathbf{a} = -\boldsymbol{\beta}_c$ gives $\boldsymbol{\beta}_c = \mathbf{0}$. Then $\pi_{ij}/\pi_{ic} = \exp(\mathbf{x}_i'\boldsymbol{\beta}_j)/\exp(0)$. If two normalised parameter sets give the same probabilities, $\mathbf{x}_i'\boldsymbol{\beta}_j = \mathbf{x}_i'\boldsymbol{\beta}_j^\dagger$ for every i, so $\mathbf{X}(\boldsymbol{\beta}_j - \boldsymbol{\beta}_j^\dagger) = \mathbf{0}$ and $\boldsymbol{\beta}_j = \boldsymbol{\beta}_j^\dagger$. For $c = 2$, $\pi_{i1} = e^{\mathbf{x}_i'\boldsymbol{\beta}_1}/(e^{\mathbf{x}_i'\boldsymbol{\beta}_1} + 1) = \Lambda(\mathbf{x}_i'\boldsymbol{\beta}_1)$. □

Corollary 7.29 (The baseline does not matter; ISLR §4.3.5)**.** With $\operatorname{rank}(\mathbf{X}) = k+1$, the maximised log-likelihood and the fitted probabilities are the same whichever category is the baseline, and the estimates with baseline h are $\mathbf{b}_j - \mathbf{b}_h$, $j = 1, \ldots, c$, where $\mathbf{b}_1, \ldots, \mathbf{b}_c$ are those with baseline c ($\mathbf{b}_c = \mathbf{0}$).

Proof. The map $(\boldsymbol{\beta}_1, \ldots, \boldsymbol{\beta}_c) \mapsto (\boldsymbol{\beta}_1 - \boldsymbol{\beta}_h, \ldots, \boldsymbol{\beta}_c - \boldsymbol{\beta}_h)$ takes the parameter sets with $\boldsymbol{\beta}_c = \mathbf{0}$ one-to-one onto those with $\boldsymbol{\beta}_h = \mathbf{0}$, its inverse subtracting the cth vector, and by Proposition 7.28 (with $\mathbf{a} = -\boldsymbol{\beta}_h$) it leaves every π_{ij} unchanged. L depends on the parameters only through the π_{ij} (Definition 7.26), so the map carries maximisers to maximisers, with the same probabilities and the same maximum. □

Proposition 7.30 (Pairs of categories; FREES §11.5.1)**.** In the generalised logit model, for categories $j \neq h$,

$$\frac{\pi_{ij}}{\pi_{ih}} = \exp\{\mathbf{x}_i'(\boldsymbol{\beta}_j - \boldsymbol{\beta}_h)\}, \qquad \mathsf{P}(y_i = j \mid y_i \in \{j, h\}) = \Lambda\big(\mathbf{x}_i'(\boldsymbol{\beta}_j - \boldsymbol{\beta}_h)\big),$$

and given $y_i \neq 1$ the probabilities of categories $2, \ldots, c$ are a generalised logit model in those categories.

Proof. The common denominator cancels in π_{ij}/π_{ih}. Then $\mathsf{P}(y_i = j \mid y_i \in \{j, h\}) = \pi_{ij}/(\pi_{ij} + \pi_{ih}) = e^u/(1+e^u)$ with $u = \mathbf{x}_i'(\boldsymbol{\beta}_j - \boldsymbol{\beta}_h)$. Given $y_i \neq 1$, the probability of $j \geqslant 2$ is $\pi_{ij}/(1-\pi_{i1}) = \exp(V_{ij})/\sum_{h \geqslant 2} \exp(V_{ih})$. □

Proposition 7.31 (The intercept-only model; FREES §11.5.1)**.** With $\mathbf{x}_i = 1$, $\boldsymbol{\beta}_j = \alpha_j$ and $\alpha_c = 0$, $\pi_j = e^{\alpha_j}/(1 + \sum_{h<c} e^{\alpha_h})$ and $\alpha_j = \ln(\pi_j/\pi_c)$. If each category occurs $n_j > 0$ times, the maximum likelihood estimate is $\hat{\pi}_j = n_j/n$, that is $\hat{\alpha}_j = \ln(n_j/n_c)$.

Proof. The first formula is Definition 7.27 with $e^{\alpha_c} = 1$, and the second is Proposition 7.28. The map $\boldsymbol{\alpha} \mapsto (\pi_1, \ldots, \pi_c)$ is a bijection of $\mathbb{R}^{c-1}$ onto the probability vectors with positive entries, its inverse being $\alpha_j = \ln(\pi_j/\pi_c)$, and $L = \sum_j n_j \ln \pi_j$. The function $\ln x - x + 1$ has derivative $1/x - 1$, positive on $(0, 1)$ and negative on $(1, \infty)$, so by the prerequisites' mean value theorem $\ln x \leqslant x - 1$ with equality only at $x = 1$. For such a vector, then,

$$\sum_j n_j \ln \pi_j - \sum_j n_j \ln \frac{n_j}{n} = \sum_j n_j \ln \frac{\pi_j}{n_j/n} \leqslant \sum_j n_j \left(\frac{n\pi_j}{n_j} - 1 \right) = n \sum_j \pi_j - n = 0,$$

with equality iff every $\pi_j = n_j/n$. □

Theorem 7.32 (Score and concavity of the generalised logit)**.** Under $\boldsymbol{\beta}_c = \mathbf{0}$ and with $\boldsymbol{\beta} =$

$(\boldsymbol{\beta}_1', \ldots, \boldsymbol{\beta}_{c-1}')'$,

$$\frac{\partial L}{\partial \boldsymbol{\beta}_j} = \sum_{i=1}^{n} \mathbf{x}_i (y_{ij} - \pi_{ij}), \qquad j < c,$$

and L is concave, strictly when $\operatorname{rank}(\mathbf{X}) = k + 1$. At a maximum likelihood estimate with an intercept, the fitted counts $\sum_i \hat{\pi}_{ij}$ equal the observed counts n_j.

Proof. With $\mathbf{v}_i = (\mathbf{x}_i'\boldsymbol{\beta}_1, \ldots, \mathbf{x}_i'\boldsymbol{\beta}_{c-1})'$ and $A(\mathbf{v}) = \ln(1 + \sum_{h<c} e^{v_h})$, $\ln \pi_{ij} = v_{ij} - A(\mathbf{v}_i)$ for $j < c$ and $\ln \pi_{ic} = -A(\mathbf{v}_i)$; as $\sum_j y_{ij} = 1$, $L = \sum_i \{\sum_{j<c} y_{ij} \mathbf{x}_i'\boldsymbol{\beta}_j - A(\mathbf{v}_i)\}$. Now $\partial A / \partial v_j = \pi_{ij}$, and the chain rule with $\partial \mathbf{v}_i / \partial \boldsymbol{\beta}_j$ acting through $\mathbf{x}_i$ gives the score. Differentiating again, $\partial^2 A / \partial v_j \partial v_h = \pi_{ij}(\delta_{jh} - \pi_{ih})$, so for $\mathbf{u} \in \mathbb{R}^{c-1}$, $\mathbf{u}' \nabla^2 A \, \mathbf{u} = \sum_{j<c} \pi_{ij} u_j^2 - (\sum_{j<c} \pi_{ij} u_j)^2 = \operatorname{var}(U)$, where U takes the value u_j with probability π_{ij}, $j < c$, and 0 with probability $\pi_{ic} > 0$; $\operatorname{var}(U) = 0$ only if U is constant, that is $\mathbf{u} = \mathbf{0}$. So A is strictly convex, and along a direction $(\mathbf{d}_1', \ldots, \mathbf{d}_{c-1}')'$ the second derivative of L is $-\sum_i \mathbf{u}_i' \nabla^2 A(\mathbf{v}_i) \mathbf{u}_i \leqslant 0$ with $\mathbf{u}_i = (\mathbf{x}_i'\mathbf{d}_1, \ldots, \mathbf{x}_i'\mathbf{d}_{c-1})'$, zero only if every $\mathbf{X}\mathbf{d}_j = \mathbf{0}$, which for $\operatorname{rank}(\mathbf{X}) = k + 1$ means $\mathbf{d} = \mathbf{0}$; the prerequisites' lemma **convexity from the derivative** gives the concavity and the strict concavity. At a maximiser the score vanishes;

its intercept coordinates say $\sum_i \hat{\pi}_{ij} = n_j$ for $j < c$, and the counts for $j = c$ agree because both sides sum to n over j. □

Example 7.15 (Job turnover; FREES §11.5.1, Table 11.6)**.** Valletta's generalised logit with baseline "no change in employment" has, for dismissal, tenure coefficient -0.221 and tenure × time trend coefficient 0.008. In 1992, time 16:

```
round(c(tenure_coefficient_1992 = -0.221 + 16 * 0.008, factor = exp(-0.221 + 16 *
  0.008),
        decline = 1 - exp(-0.221 + 16 * 0.008)), 3)
```

```
## tenure_coefficient_1992                  factor
##                  -0.093                   0.911
##                 decline
##                   0.089
```

By Proposition 7.30, an extra year of tenure in 1992 multiplies the odds of dismissal against no change by 0.911, a decline of 0.089. FREES reads this as the dismissal probability, which moves by the same factor only if the probability of no change stays fixed.

Example 7.16 (Marital status in the MEPS data)**.** A generalised logit of marital status on age and sex, with MARRIED as the baseline; the odds against being married for ten more years of age:

```
H$MARISTAT <- relevel(factor(H$MARISTAT), ref = "MARRIED")
gl <- nnet::multinom(MARISTAT ~ AGE + GENDER, data = H, trace = FALSE, reltol =
 ↪  1e-12)
round(coef(gl), 4)
```

```
##          (Intercept)     AGE  GENDER
## DIVSEP       -2.349  0.0128  0.5331
## NEVMAR        3.325 -0.1089 -0.2572
## WIDOWED      -9.567  0.1148  1.2061
```

```
round(exp(10 * coef(gl)[, "AGE"]), 3)
```

```
##  DIVSEP  NEVMAR WIDOWED
##   1.136   0.336   3.153
```

```
rbind(observed = table(H$MARISTAT), fitted = round(colSums(fitted(gl)), 2))
```

```
##          MARRIED DIVSEP NEVMAR WIDOWED
## observed    1113    251    587      49
## fitted      1113    251    587      49
```

Against being married, ten more years of age multiply the odds of never having married by 0.336, those of being widowed by 3.153, and those of being divorced or separated by 1.136. Women have larger log odds of being divorced or separated (0.5331) and widowed (1.2061), and smaller of never having married (-0.2572). The fitted category counts equal the observed ones, as Theorem 7.32 states.

5.2 The multinomial logit

Definition 7.33 (Multinomial logit; FREES §11.5.2)**.** The **multinomial** (or **conditional**) **logit** model has explanatory variables $\mathbf{x}_{ij}$ that depend on the alternative and one parameter vector $\boldsymbol{\beta}$: $V_{ij} = \mathbf{x}_{ij}'\boldsymbol{\beta}$ and $\pi_{ij} = \exp(V_{ij})/\sum_h \exp(V_{ih})$, with log-likelihood $L = \sum_i\{\sum_j y_{ij}\mathbf{x}_{ij}'\boldsymbol{\beta} - \ln\sum_h \exp(\mathbf{x}_{ih}'\boldsymbol{\beta})\}$. The **mixed logit** has $V_{ij} = \mathbf{x}_{i1j}'\boldsymbol{\beta} + \mathbf{x}_{i2}'\boldsymbol{\beta}_j$.

Proposition 7.34 (Generalised logit as a multinomial logit; FREES §11.5.2)**.** Stacking $\boldsymbol{\beta} = (\boldsymbol{\beta}_1', \ldots, \boldsymbol{\beta}_c')'$ and letting $\mathbf{x}_{ij}$ be $\mathbf{x}_i$ in the jth block of c blocks and $\mathbf{0}$ elsewhere gives $\mathbf{x}_{ij}'\boldsymbol{\beta} = \mathbf{x}_i'\boldsymbol{\beta}_j$: the generalised logit is the multinomial logit with these explanatory variables, and the mixed logit likewise.

Proof. $\mathbf{x}_{ij}'\boldsymbol{\beta} = \sum_h \mathbf{0}'\boldsymbol{\beta}_h + \mathbf{x}_i'\boldsymbol{\beta}_j$. For the mixed logit, append the blocks for $\mathbf{x}_{i2}$ to the vector $\mathbf{x}_{i1j}$. □

Proposition 7.35 (Independence of irrelevant alternatives; FREES §11.5.2)**.** In the multinomial logit model, $\ln(\pi_{ih}/\pi_{ik}) = (\mathbf{x}_{ih} - \mathbf{x}_{ik})'\boldsymbol{\beta}$, which involves no other alternative.

Proof. The common denominator cancels in $\pi_{ih}/\pi_{ik} = \exp(\mathbf{x}_{ih}'\boldsymbol{\beta})/\exp(\mathbf{x}_{ik}'\boldsymbol{\beta})$. □

Remark (FREES §11.5.2)**.** So adding or removing an alternative leaves the odds between any two others unchanged, a drawback when two alternatives are close substitutes.

Example 7.17 (Choice of health insurance; FREES §11.5.2, Tables 11.7–11.9)**.** Browne and Frees's CPS data: the odds of group insurance against none, overall and by disability, with and without the disability law; the odds ratios of Table 11.9 for the pairs not involving the baseline, computed as ratios of those against it (Proposition 7.30); and FREES's percentage readings:

```
round(c(no_law = 59.3 / 20.1, law = 57.6 / 19.9, ratio = (57.6 / 19.9) / (59.3 /
  20.1)), 3)
```

```
## no_law    law  ratio
##  2.950  2.894  0.981
```

```
round(c(nondisabled_no = 64.2 / 20.5, nondisabled_law = 63.4 / 21.2, ratio_nd =
  (63.4 / 21.2) / (64.2 / 20.5),
        disabled_no = 24.5 / 17.6, disabled_law = 21.0 / 11.4, ratio_d = (21.0 /
         11.4) / (24.5 / 17.6)), 3)
```

```
##  nondisabled_no nondisabled_law        ratio_nd
##           3.132           2.991           0.955
##     disabled_no    disabled_law         ratio_d
##           1.392           1.842           1.323
```

```
orb <- rbind(nondisabled = c(group = 0.825, nongroup = 1.053, government = 1.010),
             disabled = c(group = 1.419, nongroup = 0.953, government = 1.664))
round(cbind(group_vs_nongroup = orb[, 1] / orb[, 2], group_vs_government = orb[, 1]
  / orb[, 3],
            nongroup_vs_government = orb[, 2] / orb[, 3]), 3)
```

```
##             group_vs_nongroup group_vs_government
## nondisabled             0.783                0.817
## disabled                1.489                0.853
```

```
##             nongroup_vs_government
## nondisabled                  1.043
## disabled                     0.573
```

```
round(c(disabled_odds_change = 1.419 - 1, nondisabled_odds_change = 0.825 - 1,
  FREES_reciprocal = 1 / 0.825 - 1), 3)
```

```
##    disabled_odds_change nondisabled_odds_change
##                   0.419                  -0.175
##        FREES_reciprocal
##                   0.212
```

From the rounded percentages the odds of group insurance against none are 2.950 without the law and 2.894 with it, a ratio of 0.981, against FREES's 2.946, 2.895 and 0.983. By disability the ratios are 0.955 for the nondisabled and 1.323 for the disabled (FREES 0.952 and 1.329): an effect in the disabled subgroup that the aggregate hides. The odds ratios of Table 11.9 that do not involve the uninsured follow from the others, 0.783, 0.817 and 1.043 for the nondisabled and 1.489, 0.853 and 0.573 for the disabled, against FREES's 0.784, 0.818, 1.043, 1.490, 0.854 and 0.573. The law multiplies the disabled's odds of group insurance against none by 1.419, an increase of 0.419, and the nondisabled's by 0.825, a decrease of 0.175; FREES's "21.2% less likely" is $1/0.825 - 1 = 0.212$,

the increase of the odds when the law is not in effect.

5.3 The nested logit

Definition 7.36 (Nested logit; FREES §11.5.3)**.** For $\rho > 0$ and $S_i = \sum_{h \geqslant 2} \exp(V_{ih}/\rho)$, the **nested logit** has

$$\pi_{i1} = \frac{\exp(V_{i1})}{\exp(V_{i1}) + S_i^{\rho}}, \qquad \pi_{ij} = (1 - \pi_{i1}) \frac{\exp(V_{ij}/\rho)}{S_i}, \quad j \geqslant 2.$$

Proposition 7.37 (The nested logit; FREES §11.5.3)**.** (i) The π_{ij} are positive and sum to 1.
(ii) With $\rho = 1$ they are the probabilities of Definition 7.27 with V_{ij} in place of $\mathbf{x}_i'\boldsymbol{\beta}_j$.
(iii) For $j, h \geqslant 2$, $\pi_{ij}/\pi_{ih} = \exp\{(V_{ij} - V_{ih})/\rho\}$, while $\pi_{i1}/\pi_{ij} = \exp(V_{i1} - V_{ij}/\rho)\, S_i^{1-\rho}$, which depends on every V_{ih}, $h \geqslant 2$, $h \neq j$, unless $\rho = 1$.

Proof. (i) $0 < \pi_{i1} < 1$, and $\sum_{j \geqslant 2} \pi_{ij} = (1 - \pi_{i1}) S_i / S_i$. (ii) With $\rho = 1$, $S_i = \sum_{h \geqslant 2} e^{V_{ih}}$, so $\pi_{i1} = e^{V_{i1}} / \sum_h e^{V_{ih}}$ and $1 - \pi_{i1} = S_i / \sum_h e^{V_{ih}}$, whence $\pi_{ij} = e^{V_{ij}} / \sum_h e^{V_{ih}}$. (iii) The first ratio is immediate. As $1 - \pi_{i1} = S_i^{\rho} / \{\exp(V_{i1}) + S_i^{\rho}\}$, $\pi_{i1}/\pi_{ij} = \exp(V_{i1}) S_i / \{S_i^{\rho} \exp(V_{ij}/\rho)\}$, and S_i is

strictly increasing in each V_{ih}, so $S_i^{1-\rho}$ is strictly increasing in each V_{ih} when $\rho < 1$ and strictly decreasing when $\rho > 1$. □

6 Ordinal Dependent Variables

FREES §11.6. SRM learning outcomes 2(a), 2(c).

6.1 The cumulative logit

Definition 7.38 (Proportional odds model; FREES §11.6.1)**.** An **ordinal** response takes the ordered values $1 < \cdots < c$. With $\alpha_1 < \cdots < \alpha_{c-1}$ and $\mathbf{x}_i$ now without the intercept, the **cumulative logit** or **proportional odds** model is

$$\operatorname{logit} \mathsf{P}(y_i \leqslant j) = \alpha_j + \mathbf{x}_i'\boldsymbol{\beta}, \qquad j = 1, \ldots, c-1,$$

and the α_j are the **cut points**.

Proposition 7.39 (Ordered cut points; FREES §11.6.1)**.** With $\alpha_0 = -\infty$ and $\alpha_c = +\infty$, the model gives $\pi_{ij} = \Lambda(\alpha_j + \mathbf{x}_i'\boldsymbol{\beta}) - \Lambda(\alpha_{j-1} + \mathbf{x}_i'\boldsymbol{\beta})$, and these are positive for every $\mathbf{x}_i$ iff $\alpha_1 < \cdots < \alpha_{c-1}$; with $\alpha_{j-1} = \alpha_j$ category j has probability 0.

Proof. $\pi_{ij} = \mathsf{P}(y_i \leqslant j) - \mathsf{P}(y_i \leqslant j-1)$, and applying Λ to the model gives the difference. Λ is strictly increasing by Proposition 7.8, so the difference is positive iff $\alpha_j > \alpha_{j-1}$, whatever $\mathbf{x}_i'\boldsymbol{\beta}$, which is added to both. □

Theorem 7.40 (Latent variable and the cumulative probit; FREES §11.6.2)**.** Let $y_i^* = -\mathbf{x}_i'\boldsymbol{\beta} + \varepsilon_i$ and $y_i = j$ when $\alpha_{j-1} < y_i^* \leqslant \alpha_j$. If ε_i has distribution function Λ, the model of Definition 7.38 holds; if ε_i is standard normal, $\mathsf{P}(y_i \leqslant j) = \Phi(\alpha_j + \mathbf{x}_i'\boldsymbol{\beta})$, the **cumulative probit** model.

Proof. $\mathsf{P}(y_i \leqslant j) = \mathsf{P}(y_i^* \leqslant \alpha_j) = \mathsf{P}(\varepsilon_i \leqslant \alpha_j + \mathbf{x}_i'\boldsymbol{\beta})$, which is $\Lambda(\alpha_j + \mathbf{x}_i'\boldsymbol{\beta})$ or $\Phi(\alpha_j + \mathbf{x}_i'\boldsymbol{\beta})$, and $\operatorname{logit}\Lambda(z) = z$. □

Remark (Signs; FREES §11.6.2)**.** FREES takes $y_i^* - \mathbf{x}_i'\boldsymbol{\beta}$ logistic, which gives $\operatorname{logit}\mathsf{P}(y_i \leqslant j) = \alpha_j - \mathbf{x}_i'\boldsymbol{\beta}$: its equation (11.12) with $-\boldsymbol{\beta}$ in place of $\boldsymbol{\beta}$. R's `MASS::polr` fits $\operatorname{logit}\mathsf{P}(y_i \leqslant j) =$

$\zeta_j - \mathbf{x}_i'\boldsymbol{\gamma}$, so $\zeta_j = \alpha_j$ and $\boldsymbol{\gamma} = -\boldsymbol{\beta}$. The conventions give the same probabilities and opposite signs: a positive γ_l moves the response towards the higher categories.

Theorem 7.41 (Proportional odds; FREES §11.6.1)**.** In the model of Definition 7.38, raising x_{il} by one unit, the other variables fixed, multiplies the cumulative odds $\mathsf{P}(y_i \leqslant j)/\,\mathsf{P}(y_i > j)$ by e^{β_l} for every j, and $\beta_l = \partial \ln \text{odds}(y_i \leqslant j)/\partial x_{il}$.

Proof. $\ln \text{odds}(y_i \leqslant j) = \alpha_j + \mathbf{x}_i'\boldsymbol{\beta}$ increases by β_l, whatever α_j, and its derivative in x_{il} is β_l. □

Corollary 7.42 (Dichotomising the response)**.** For each $j < c$, $\mathbb{1}\{y_i \leqslant j\}$ follows the logit model with intercept α_j and slopes $\boldsymbol{\beta}$, the same slopes for every j.

Proof. $\mathsf{P}(\mathbb{1}\{y_i \leqslant j\} = 1) = \Lambda(\alpha_j + \mathbf{x}_i'\boldsymbol{\beta})$, which is Definition 7.6 with the logit and linear predictor $\alpha_j + \mathbf{x}_i'\boldsymbol{\beta}$. □

Remark. So the proportional odds restriction can be checked by fitting the $c - 1$ dichotomised logit models: their slope vectors estimate the same $\boldsymbol{\beta}$.

Example 7.18 (Health plan choice; FREES §11.6.1, Table 11.10)**.** Pauly and Herring's cumulative logit of plan type, ordered from the most restrictive (HMO) to the least (fee for service), on $n = 34{,}486$ persons, reports the odds ratio 1.064 for women and the cut points as odds ratios 0.769, 1.406 and 12.089:

```
round(c(alpha = log(c(0.769, 1.406, 12.089)), female = log(1.064)), 3)
```

```
## alpha1 alpha2 alpha3 female
## -0.263  0.341  2.492  0.062
```

The cut points are -0.263, 0.341 and 2.492, increasing as Proposition 7.39 requires, and the coefficient for women is $\ln 1.064 = 0.062$. FREES reads 1.064 as women being more likely to choose less restrictive plans. Under equation (11.12) an odds ratio above 1 multiplies the odds of the more restrictive categories $y \leqslant j$, so the reading holds in the `polr` convention of the remark on signs, where the reported ratio multiplies the odds of $y > j$.

Example 7.19 (Self-rated physical health in the MEPS data)**.** `PHSTAT` ordered from excellent to poor, on age, sex, activity limitation, insurance and income, by the cumulative logit and the cumulative probit, with $\boldsymbol{\beta} = -\boldsymbol{\gamma}$ from `polr`:

```r
H$PH <- factor(H$PHSTAT, levels = c("EXCE", "VGOO", "GOOD", "FAIR", "POOR"), ordered
 ↪  = TRUE)
f_ph <- PH ~ AGE + GENDER + ANYLIMIT + insure + INCOME
po <- MASS::polr(f_ph, data = H, Hess = TRUE); pp <- MASS::polr(f_ph, data = H,
 ↪  method = "probit", Hess = TRUE)
round(rbind(beta_logit = -coef(po), odds_ratio_higher = exp(coef(po)), beta_probit =
 ↪  -coef(pp),
            ratio = coef(po) / coef(pp)), 3)
```

```
##                     AGE GENDER ANYLIMIT insure INCOMEHINCOME
## beta_logit       -0.023 -0.259   -1.197  0.228         1.339
## odds_ratio_higher  1.023  1.295    3.309  0.796         0.262
## beta_probit      -0.014 -0.158   -0.705  0.129         0.780
## ratio             1.630  1.640    1.698  1.765         1.716
##                  INCOMELINCOME INCOMEMINCOME INCOMENPOOR
## beta_logit              0.624         0.760       0.073
## odds_ratio_higher       0.536         0.468       0.930
## beta_probit             0.352         0.437       0.017
## ratio                   1.775         1.737       4.187
```

```r
round(rbind(alpha_logit = po$zeta, alpha_probit = pp$zeta), 3)
```

```
##              EXCE|VGOO VGOO|GOOD GOOD|FAIR FAIR|POOR
## alpha_logit     -0.898     0.587     2.399     3.929
## alpha_probit    -0.493     0.405     1.471     2.287
```

```
dich <- sapply(1:4, function(j) coef(glm(I(as.integer(PH) <= j) ~ AGE + GENDER +
  ANYLIMIT + insure + INCOME,
                                         family = binomial, data = H))[-1])
colnames(dich) <- paste0("y_le_", 1:4); round(cbind(dich, polr = -coef(po)), 3)
```

```
##                y_le_1 y_le_2 y_le_3 y_le_4   polr
## AGE            -0.022 -0.018 -0.033 -0.046 -0.023
## GENDER         -0.362 -0.249 -0.081 -0.370 -0.259
## ANYLIMIT       -0.898 -1.080 -1.437 -2.110 -1.197
## insure          0.270  0.340  0.114 -0.286  0.228
## INCOMEHINCOME   0.667  1.280  2.036  2.311  1.339
## INCOMELINCOME  -0.008  0.603  1.018  1.154  0.624
## INCOMEMINCOME   0.134  0.631  1.318  1.867  0.760
## INCOMENPOOR    -0.585  0.006  0.286  0.704  0.073
```

In the convention of Definition 7.38 (FREES's (11.12)) a negative coefficient moves the response towards poorer health: age (-0.023 per year), women (-0.259) and an activity limitation (-1.197,

which multiplies the odds of a poorer category by 3.309) do so, while insurance (0.228) and every income level above poor act the other way, the highest most (1.339). The cut points increase from -0.898 to 3.929. The logit coefficients are 1.630 to 1.775 times the probit ones, except for the near-poor level, where both are near zero (0.073 and 0.017). The dichotomised logits drift: the activity limitation coefficient goes from -0.898 at $y \leqslant 1$ to -2.110 at $y \leqslant 4$, and high income from 0.667 to 2.311, so by Corollary 7.42 the proportional odds restriction is doubtful for these variables.

7 Classification

ISLR §4.3 and §4.3.3 (PA reading list): classification by a threshold on the fitted probability, 1/2 or a lower one for a conservative lender. Beyond the syllabus: the confusion matrix, sensitivity, specificity, the ROC curve and the AUC (ISLR §4.4.2), and FREES §11, which fits probabilities and does not classify.

Definition 7.43 (Classifier and confusion counts; ISLR §4.3, §4.4.2)**.** With fitted probabilities $\hat{p}_i$ and a threshold $t \in (0,1)$, the classifier predicts $\hat{y}_i = \mathbb{1}\{\hat{p}_i > t\}$. The counts TP, FN, FP, TN of $(\hat{y}_i, y_i)$ equal to $(1,1), (0,1), (1,0), (0,0)$ (the confusion matrix, beyond the syllabus) give the

sensitivity TP/(TP+FN), the **specificity** TN/(TN+FP) (these two beyond the syllabus) and the **accuracy** (TP+TN)/n.

Theorem 7.44 (The cost-optimal threshold)**.** Let $(\mathbf{X}, y)$ have $\mathsf{P}(y = 1 \mid \mathbf{X}) = p(\mathbf{X})$, and let predicting 0 when $y = 1$ cost $c_{01} > 0$ and predicting 1 when $y = 0$ cost $c_{10} > 0$. Among classifiers $h(\mathbf{X}) \in \{0, 1\}$, the expected cost is minimised by $h^*(\mathbf{X}) = \mathbb{1}\{p(\mathbf{X}) > c_{10}/(c_{10} + c_{01})\}$.

Proof. By the defining property of the prerequisites' theorem **conditional expectation in L^2**, with W the indicators $\mathbb{1}\{h(\mathbf{X}) = 0\}$ and $\mathbb{1}\{h(\mathbf{X}) = 1\}$, the expected cost is $\mathsf{E}[c_{01}p(\mathbf{X})\,\mathbb{1}\{h(\mathbf{X}) = 0\} + c_{10}\{1 - p(\mathbf{X})\}\,\mathbb{1}\{h(\mathbf{X}) = 1\}]$. For each value of $\mathbf{X}$ the integrand is smaller with $h = 1$ iff $c_{10}\{1 - p(\mathbf{X})\} < c_{01}p(\mathbf{X})$, that is $p(\mathbf{X}) > c_{10}/(c_{10} + c_{01})$, and h^* makes the integrand pointwise smallest. □

Corollary 7.45. With equal costs the threshold is 1/2 and h^* is Unit 1's Bayes classifier (definition **Bayes classifier**), which minimises the error rate (proposition **optimality of the Bayes classifier**).

Proof. $c_{10}/(c_{10}+c_{01}) = 1/2$, and with unit costs the expected cost is the error rate. □

Definition 7.46 (ROC curve and AUC (beyond the syllabus))**.** For scores s_i of n_1 observations with $y_i = 1$ and n_0 with $y_i = 0$, let $\mathrm{TPR}(t) = \#\{i : y_i = 1, s_i \geqslant t\}/n_1$ and $\mathrm{FPR}(t) = \#\{i : y_i = 0, s_i \geqslant t\}/n_0$. The **ROC curve** joins by segments the point $(0,0)$ and the points $(\mathrm{FPR}(t), \mathrm{TPR}(t))$ as t decreases through the distinct scores, the last being $(1,1)$, and the **AUC** is the area under it.

Theorem 7.47 (The area counts ordered pairs (beyond the syllabus))**.** $\mathrm{AUC} = \big[\#\{(i,j) : y_i = 1, y_j = 0, s_i > s_j\} + \frac{1}{2}\#\{(i,j) : y_i = 1, y_j = 0, s_i = s_j\}\big]/(n_1 n_0)$.

Proof. As t decreases to a score value v held by a_v observations with $y = 0$ and b_v with $y = 1$, the curve moves by $(a_v/n_0, b_v/n_1)$ from the previous point, of height $\#\{i : y_i = 1, s_i > v\}/n_1$, and the trapezoid under this segment has area $(a_v/n_0)\{\#\{i : y_i = 1, s_i > v\} + b_v/2\}/n_1$. Summing over v, the first terms count the pairs with $s_i > s_j$ and the second the ties, each tie counted with weight $1/2$. □

Corollary 7.48. A strictly increasing transformation of the scores leaves the ROC curve and the AUC unchanged, and so does any change of n_1 and n_0 that keeps the two within-class score distributions.

Proof. A strictly increasing map preserves every comparison $s_i > s_j$ and $s_i = s_j$ and the order of the thresholds. TPR and FPR are proportions within the two classes. □

Example 7.20 (Classifying hospitalisation)**.** The reduced logit fit of Example 7.13 as a classifier at $t = 1/2$ and at the sample proportion of hospitalised persons, against the rule that predicts 0 always; and the AUC (beyond the syllabus) of Theorem 7.47, of the trapezoids, and of the linear predictor:

```
ph <- fitted(red); yv <- H$POSEXP
conf <- sapply(c(0.5, mean(yv)), function(t) { yh <- as.numeric(ph > t)
  c(t = t, TP = sum(yh == 1 & yv == 1), FN = sum(yh == 0 & yv == 1),
    FP = sum(yh == 1 & yv == 0), TN = sum(yh == 0 & yv == 0),
    sensitivity = mean(yh[yv == 1]), specificity = mean(1 - yh[yv == 0]),
    accuracy = mean(yh == yv)) })
```

```r
colnames(conf) <- c("t_half", "t_rate"); round(conf, 4); round(c(accuracy_of_always_0 = mean(yv == 0)), 4)
```

```
##                  t_half    t_rate
## t                0.5000    0.0785
## TP               5.0000  109.0000
## FN             152.0000   48.0000
## FP               8.0000  565.0000
## TN            1835.0000 1278.0000
## sensitivity      0.0318    0.6943
## specificity      0.9957    0.6934
## accuracy         0.9200    0.6935
## accuracy_of_always_0
##                  0.9215
```

```r
pair_auc <- function(s) { d <- outer(s[yv == 1], s[yv == 0], "-"); mean((d > 0) + 0.5 * (d == 0)) }
thr <- sort(unique(ph), decreasing = TRUE)
tpr <- c(0, sapply(thr, function(t) mean(ph[yv == 1] >= t))); fpr <- c(0, sapply(thr, function(t) mean(ph[yv == 0] >= t)))
```

```
round(c(pairs = pair_auc(ph), trapezoids = sum(diff(fpr) * (head(tpr, -1) +
 ↪  tail(tpr, -1)) / 2),
       linear_predictor = pair_auc(qlogis(ph))), 4)
```

```
##           pairs      trapezoids linear_predictor
##          0.7588          0.7588          0.7588
```

```
op <- par(mar = c(3.2, 3.8, 0.6, 0.6), mgp = c(2.1, 0.6, 0))
plot(fpr, tpr, type = "l", xlab = "false positive rate",
     ylab = "true positive rate"); abline(0, 1, lty = 3); par(op)
```

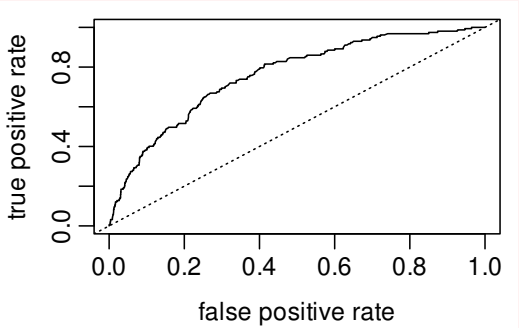


Figure 7.4: ROC curve of the reduced logit model for hospitalisation, with the diagonal of a score unrelated to the response (dotted).

At $t = 1/2$ the classifier finds 5 hospitalised persons and misses 152, a sensitivity of 0.0318, and its accuracy 0.9200 is below the 0.9215 of predicting 0 always. At the sample proportion 0.0785 the sensitivity is 0.6943 and the specificity 0.6934. The AUC is 0.7588 by pairs and by trapezoids, as Theorem 7.47 states, and for the linear predictor, as Corollary 7.48 states.

8 Computation in R

FREES §11.4 and its chapter script.

8.1 FREES's chapter script

FREES's script tabulates the percentages of Example 7.12 with `Hmisc`, fits the logit on sex with `Design::lrm`, whose third statistic is the model likelihood ratio $2\{L(\mathbf{b}) - L_0\}$, and compares the reduced and full logit models with `anova`; its full model omits `MNHPOOR`, and here the full model of Example 7.13 is used:

```
1 c(model_LR_sex = g1$null.deviance - g1$deviance, df = 1)
```

```
## model_LR_sex           df
##        26.38         1.00
```

```
anova(red, full, test = "Chisq")
```

```
## Analysis of Deviance Table
##
## Model 1: POSEXP ~ GENDER + RACE + REGION + EDUC + PHSTAT + ANYLIMIT +
##     INCOME + insure
## Model 2: POSEXP ~ AGE + GENDER + RACE + REGION + EDUC + PHSTAT + MNHPOOR +
##     ANYLIMIT + INCOME + insure
##   Resid. Df Resid. Dev Df Deviance Pr(>Chi)
## 1      1980        978
## 2      1978        977  2    0.166     0.92
```

The model likelihood ratio for sex alone is 26.38 on 1 degree of freedom, and **anova** repeats the statistic 0.166 of Example 7.13, with p-value 0.92.

8.2 Formulas against package output

The score at the fit, the standard errors from $\mathbf{X}'\mathbf{W}\mathbf{X}$, Fisher scoring (Theorem 7.18) for both links, the log-likelihood and AIC, and the Wald, z^2 and likelihood ratio statistics for sex (Corollary 7.21):

```
X <- model.matrix(red); b <- coef(red); pr <- drop(plogis(X %*% b))
Info <- crossprod(X, pr * (1 - pr) * X)
```

```r
scoring <- function(X, y, Fn, fn) {
  bb <- numeric(ncol(X))
  repeat { eta <- drop(X %*% bb); p <- Fn(eta); f <- fn(eta)
    w <- f^2 / (p * (1 - p)); z <- eta + (y - p) / f
    bn <- drop(solve(crossprod(X, w * X), crossprod(X, w * z)))
    if (max(abs(bn - bb)) < 1e-10) return(bn); bb <- bn } }
signif(c(max_abs_score = max(abs(crossprod(X, yv - pr))),
  se_manual_vs_glm = max(abs(sqrt(diag(solve(Info))) - coef(summary(red))[, 2])),
  scoring_logit_vs_glm = max(abs(scoring(X, yv, plogis, dlogis) - b)),
  scoring_probit_vs_glm = max(abs(scoring(X, yv, pnorm, dnorm) - coef(pro))),
  loglik_manual_minus_glm = sum(yv * log(pr) + (1 - yv) * log(1 - pr)) -
   ↪  as.numeric(logLik(red)),
  AIC_manual_minus_glm = -2 * as.numeric(logLik(red)) + 2 * ncol(X) - AIC(red)), 3)
```

```
##           max_abs_score         se_manual_vs_glm
##                1.03e-09                1.49e-06
##    scoring_logit_vs_glm   scoring_probit_vs_glm
##                8.07e-11                1.18e-05
## loglik_manual_minus_glm    AIC_manual_minus_glm
##                0.00e+00                0.00e+00
```

```
j <- which(names(b) == "GENDER"); r0 <- glm(update(f_red, . ~ . - GENDER), family =
↪  binomial, data = H)
round(c(Wald = b[[j]]^2 / solve(Info)[j, j], z_squared = coef(summary(red))[j, 3]^2,
        LRT = 2 * as.numeric(logLik(red) - logLik(r0))), 4)
```

```
##      Wald z_squared       LRT
##     14.49     14.49     15.42
```

The score vanishes at the fit to 1.03×10^{-9}, the standard errors from $\mathbf{X}'\mathbf{W}\mathbf{X}$ agree with `glm`'s to 1.49×10^{-6}, Fisher scoring reproduces the logit fit to 8.07×10^{-11} and the probit fit to 1.18×10^{-5}, and the log-likelihood and AIC agree exactly. For sex, $W = z^2 = 14.49$, and the likelihood ratio statistic, 15.42, differs from it in this finite sample. Newton–Raphson for the generalised logit of Example 7.16, with $-\nabla^2 L$, whose (a,h) block $\sum_i \pi_{ia}(\delta_{ah} - \pi_{ih})\mathbf{x}_i\mathbf{x}_i'$ follows from $\partial^2 A/\partial v_a \partial v_h$ in the proof of Theorem 7.32, against `nnet::multinom`; the pair `NEVMAR`–`MARRIED` fitted alone (Proposition 7.30); and the cumulative logit likelihood of Example 7.19 against `polr`:

```
Xm <- model.matrix(~ AGE + GENDER, data = H); Ym <- model.matrix(~ MARISTAT - 1, data =
↪  H)
gl_newton <- function(X, Y) { q <- ncol(X); m <- ncol(Y) - 1; B <- matrix(0, q, m)
```

```
  repeat { V <- cbind(0, X %*% B); P <- exp(V) / rowSums(exp(V)); g <-
   ↪  as.vector(crossprod(X, Y[, -1] - P[, -1]))
    I <- matrix(0, q * m, q * m)
    for (a in 1:m) for (h in 1:m) I[(a - 1) * q + 1:q, (h - 1) * q + 1:q] <-
     ↪  crossprod(X, P[, a + 1] * ((a == h) - P[, h + 1]) * X)
    s <- solve(I, g); B <- B + matrix(s, q, m); if (max(abs(s)) < 1e-12) return(t(B)) }
     ↪  }
Bn <- gl_newton(Xm, Ym); V <- cbind(0, Xm %*% t(Bn)); Pm <- exp(V) / rowSums(exp(V))
signif(c(newton_vs_multinom = max(abs(Bn - coef(gl))), probabilities_vs_multinom =
 ↪  max(abs(Pm - fitted(gl))),
  loglik_newton_minus_multinom = sum(Ym * log(Pm)) - as.numeric(logLik(gl))), 3)
```

```
##           newton_vs_multinom    probabilities_vs_multinom
##                     4.58e-07                     4.28e-08
## loglik_newton_minus_multinom
##                     6.82e-13
```

```
sub <- droplevels(subset(H, MARISTAT %in% c("MARRIED", "NEVMAR")))
round(rbind(generalised_logit = Bn[which(levels(H$MARISTAT) == "NEVMAR") - 1, ],
            two_category_logit = coef(glm(I(MARISTAT == "NEVMAR") ~ AGE + GENDER,
             ↪  family = binomial, data = sub))), 3)
```

```
##                     (Intercept)    AGE GENDER
## generalised_logit         3.326 -0.109 -0.257
## two_category_logit        3.173 -0.105 -0.250
```

```
Xo <- model.matrix(f_ph, data = H)[, -1]; cum <- sapply(po$zeta, function(a) plogis(a -
 ↪  drop(Xo %*% coef(po))))
Po <- cbind(cum, 1) - cbind(0, cum)
signif(c(loglik_manual_minus_polr = sum(log(Po[cbind(seq_len(nrow(Po)),
 ↪  as.integer(H$PH))])) - as.numeric(logLik(po)),
  probabilities_vs_predict = max(abs(Po - predict(po, type = "probs")))), 3)
```

```
## loglik_manual_minus_polr probabilities_vs_predict
##                        0                        0
```

Newton–Raphson agrees with **multinom** to 4.58×10^{-7} in the coefficients and 4.28×10^{-8} in the probabilities. The pair fitted alone gives 3.173, −0.105 and −0.250 against 3.326, −0.109 and −0.257 from the generalised logit, two estimates of the same vector under the model. The cumulative logit likelihood and probabilities agree with **polr** exactly.

8.3 The ISLR lab (beyond the syllabus)

Beyond the syllabus: ISLR §4.7.1–4.7.2, the chapter's lab, which neither reading list includes.

Example 7.21 (Stock market directions; ISLR §4.7.1–4.7.2)**.** `Smarket`: 1250 daily S&P 500 returns, 2001–2005, with the five previous returns `Lag1`–`Lag5`, `Volume` and `Direction` (`Up` or `Down`). The logit on all six, its classifications at threshold 1/2, the fit on 2001–2004 tested on 2005, and the model with `Lag1` and `Lag2` only:

```
Sm <- Smarket; CS <- cor(Sm[, -9]); round(CS["Year", "Volume"], 3)
```

```
## [1] 0.539
```

```
diag(CS) <- 0; CS["Year", "Volume"] <- CS["Volume", "Year"] <- 0;
 ↪  round(max(abs(CS)), 3)
```

```
## [1] 0.048
```

```
fS6 <- Direction ~ Lag1 + Lag2 + Lag3 + Lag4 + Lag5 + Volume
gS <- glm(fS6, data = Sm, family = binomial)
round(coef(summary(gS)), 4)
```

```
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept)  -0.1260     0.2407 -0.5234   0.6007
## Lag1         -0.0731     0.0502 -1.4566   0.1452
## Lag2         -0.0423     0.0501 -0.8446   0.3983
## Lag3          0.0111     0.0499  0.2220   0.8243
## Lag4          0.0094     0.0500  0.1873   0.8514
## Lag5          0.0103     0.0495  0.2083   0.8350
## Volume        0.1354     0.1584  0.8553   0.3924
```

```
cls <- function(p) ifelse(p > 0.5, "Up", "Down")
pS <- cls(fitted(gS)); table(pS, Sm$Direction); mean(pS == Sm$Direction)
```

```
##
## pS     Down  Up
##   Down  145 141
##   Up    457 507
## [1] 0.5216
```

```
trS <- Sm$Year < 2005; d05 <- Sm$Direction[!trS]
gS2 <- glm(fS6, data = Sm, family = binomial, subset = trS)
```

```
p2 <- cls(predict(gS2, Sm[!trS, ], type = "response")); table(p2, d05); mean(p2 !=
 ↪  d05)
```

```
##       d05
## p2     Down Up
##   Down   77 97
##   Up     34 44
## [1] 0.5198
```

```
gS3 <- glm(Direction ~ Lag1 + Lag2, data = Sm, family = binomial, subset = trS)
p3 <- cls(predict(gS3, Sm[!trS, ], type = "response")); table(p3, d05)
```

```
##       d05
## p3     Down  Up
##   Down   35  35
##   Up     76 106
```

```
round(c(accuracy = mean(p3 == d05), always_up = mean(d05 == "Up"),
        up_when_predicted_up = sum(p3 == "Up" & d05 == "Up") / sum(p3 == "Up")), 4)
```

```
##             accuracy            always_up
##               0.5595               0.5595
```

```
## up_when_predicted_up
##               0.5824
```

```
round(predict(gS3, data.frame(Lag1 = c(1.2, 1.5), Lag2 = c(1.1, -0.8)), type =
  "response"), 4)
```

```
##      1      2
## 0.4791 0.4961
```

The only sizeable correlation among the variables is **Year** with **Volume**, 0.539; every other is at most 0.048 in absolute value. No coefficient is significant, the smallest p-value being **Lag1**'s 0.145. The training accuracy is 52.2%; on 2005, after fitting to 2001–2004, the error rate is 52.0%. With **Lag1** and **Lag2** the 2005 accuracy is 56.0%, exactly that of predicting **Up** every day, and 58.2% of the days predicted **Up** are **Up**. The predicted probabilities of **Up** at the two new days are 0.4791 and 0.4961. The training accuracy is the training error rate of Unit 1 subtracted from 1, an optimistic estimate of the test accuracy, which the 2005 hold-out estimates.

9 Exercises

Exercise 7.1 (FREES exercise 11.1)**.** Let y^* have the distribution function $F(y) = e^y/(1+e^y)$. (a) Find its density. (b) Find its mean μ_y. (c) Find its standard deviation σ_y. (d) Find the density of $y^{**} = (y^* - \mu_y)/\sigma_y$. (e) Plot that density with the standard normal density.

Solution. (a) $F = \Lambda$, so the density is $F'(y) = e^y/(1+e^y)^2 = \Lambda(y)\{1-\Lambda(y)\}$ by Proposition 7.8. (b) and (c) $\mu_y = 0$ and $\sigma_y = \pi/\sqrt{3}$, with $\pi = 3.14159\ldots$ the number, by Proposition 7.12(ii). (d) $\mathsf{P}(y^{**} \leqslant a) = F(\sigma_y a)$, so y^{**} has density $\sigma_y F'(\sigma_y a) = \sigma_y e^{\sigma_y a}/(1+e^{\sigma_y a})^2$. (e):

```
s_y <- pi / sqrt(3); a <- seq(-4, 4, length = 801)
round(c(sigma_y = s_y, max_density_difference = max(abs(s_y * dlogis(s_y * a) -
 ↪  dnorm(a)))), 4)
```

```
##               sigma_y max_density_difference
##                1.8138                 0.0545
```

```
op <- par(mar = c(3.2, 3.8, 0.6, 0.6), mgp = c(2.1, 0.6, 0))
plot(a, dnorm(a), type = "l", xlab = "y", ylab = "density", ylim = c(0, 0.47))
lines(a, s_y * dlogis(s_y * a), lty = 2); par(op)
```

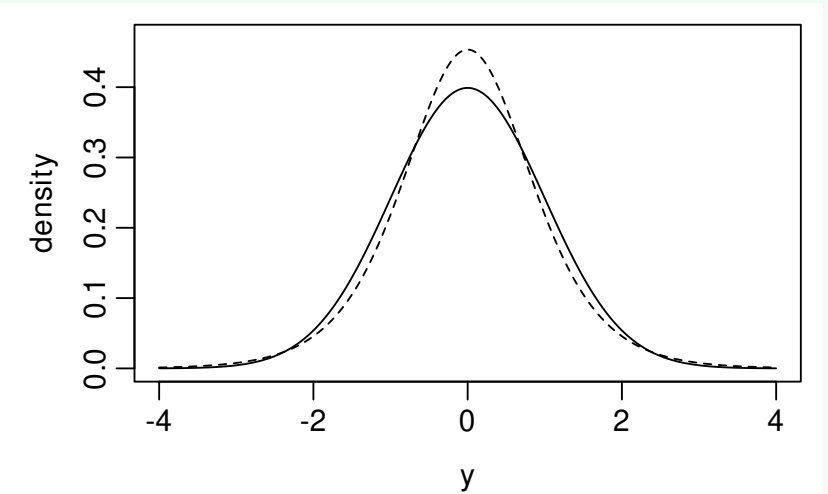


Figure 7.5: The density of y^{**} (dashed) and the standard normal density (solid).

$\sigma_y = 1.8138$, and the two densities differ by at most 0.0545; the logistic density is the higher at the centre and in the tails. ■

Exercise 7.2 (FREES exercise 11.2)**.** Let $y_i^* = \mathbf{x}_i'\boldsymbol{\beta} + \varepsilon_i^*$ with $\varepsilon_i^* \sim N(0, \sigma^2)$ and $y_i = \mathbb{1}\{y_i^* > 0\}$. Show that $\mathsf{P}(y_i = 1 \mid \mathbf{x}_i) = \Phi(\mathbf{x}_i'\boldsymbol{\beta}/\sigma)$.

Solution. $\varepsilon_i^* = \sigma Z$ with Z standard normal, so $\mathsf{P}(y_i = 1) = \mathsf{P}(Z > -\mathbf{x}_i'\boldsymbol{\beta}/\sigma) = 1 - \Phi(-\mathbf{x}_i'\boldsymbol{\beta}/\sigma) = \Phi(\mathbf{x}_i'\boldsymbol{\beta}/\sigma)$, the normal density being even; this is Theorem 7.9 with $\pi = \Phi$. ■

Exercise 7.3 (FREES exercise 11.3)**.** Under the random utility interpretation with $V_{i2} = 0$, $V_{i1} = \mathbf{x}_i'\boldsymbol{\beta}$ and errors with $\mathsf{P}(\varepsilon_{ij} < a) = \exp(-e^{-a})$, show that $\pi_i = \mathsf{P}(\varepsilon_{i2} - \varepsilon_{i1} < V_{i1} - V_{i2}) = 1/\{1 + \exp(-\mathbf{x}_i'\boldsymbol{\beta})\}$.

Solution. The exercise leaves the joint law of the two errors open; with ε_{i1} and ε_{i2} independent, Proposition 7.10 gives $\mathsf{P}(\varepsilon_{i2} - \varepsilon_{i1} \leqslant t) = \Lambda(t)$ for every t, and Λ is continuous, so $\pi_i = \Lambda(\mathbf{x}_i'\boldsymbol{\beta}) = 1/\{1 + \exp(-\mathbf{x}_i'\boldsymbol{\beta})\}$. ■

Exercise 7.4 (FREES exercise 11.4)**.** (a) For an i.i.d. Bernoulli sample with mean π, show that the maximum likelihood estimator of π is $\bar{y}$. (b) For independent Bernoulli samples of sizes n_1 and n_2 with means π_1 and π_2: (i) show that the maximum likelihood estimator of $\pi_2 - \pi_1$ is $\bar{y}_2 - \bar{y}_1$; (ii) find its variance. (c) In the logit model with an intercept and a 0/1 variable x, n_1 observations having $x = 0$ and mean $\bar{y}_1$, and n_2 having $x = 1$ and mean $\bar{y}_2$: (i) show that $\bar{y}_1 = \pi(b_0)$ and $\bar{y}_2 = \pi(b_0 + b_1)$; (ii) show that $b_1 = \pi^{-1}(\bar{y}_2) - \pi^{-1}(\bar{y}_1)$; (iii) confirm that

$\mathcal{I}(\beta_0,\beta_1) = n_1\pi_1(1-\pi_1)\left(\begin{smallmatrix}1&0\\0&0\end{smallmatrix}\right) + n_2\pi_2(1-\pi_2)\left(\begin{smallmatrix}1&1\\1&1\end{smallmatrix}\right)$ with $\pi_1 = \pi(\beta_0)$ and $\pi_2 = \pi(\beta_0+\beta_1)$; (iv) find the large-sample variance of b_1.

Solution. (a) $L(\pi) = n\bar{y}\ln\pi + n(1-\bar{y})\ln(1-\pi)$. If $0<\bar{y}<1$, $L'(\pi) = n(\bar{y}-\pi)/\{\pi(1-\pi)\}$ is positive below $\bar{y}$ and negative above, so $\bar{y}$ is the maximiser by the prerequisites' mean value theorem; if $\bar{y}=0$ or 1, L is monotone on $[0,1]$ with maximum at $\pi=\bar{y}$.
(b)(i) The log-likelihood is $L_1(\pi_1)+L_2(\pi_2)$, maximised at $(\bar{y}_1,\bar{y}_2)$ by (a), and $(\pi_1,\pi_2)\mapsto(\pi_1,\pi_2-\pi_1)$ is a bijection, so the maximiser in the new parameters is $(\bar{y}_1,\bar{y}_2-\bar{y}_1)$. (ii) By independence and Proposition 7.2, $\operatorname{var}(\bar{y}_2-\bar{y}_1) = \pi_1(1-\pi_1)/n_1 + \pi_2(1-\pi_2)/n_2$.
(c)(i) and (ii) are Corollary 7.16, for $0<\bar{y}_1,\bar{y}_2<1$. (iii) By Theorem 7.14, $\mathcal{I} = \sum_i \pi_i(1-\pi_i)\mathbf{x}_i\mathbf{x}_i'$, with $\mathbf{x}_i\mathbf{x}_i'$ equal to $\left(\begin{smallmatrix}1&0\\0&0\end{smallmatrix}\right)$ for the n_1 observations with $x_i=0$ and to $\left(\begin{smallmatrix}1&1\\1&1\end{smallmatrix}\right)$ for the n_2 with $x_i=1$.
(iv) With $u = n_1\pi_1(1-\pi_1)$ and $v = n_2\pi_2(1-\pi_2)$, $\mathcal{I} = \left(\begin{smallmatrix}u+v&v\\v&v\end{smallmatrix}\right)$ has the inverse $(uv)^{-1}\left(\begin{smallmatrix}v&-v\\-v&u+v\end{smallmatrix}\right)$, as multiplication verifies, with lower right entry $(u+v)/(uv) = 1/u+1/v$; by Theorem 7.19 the large-sample variance of b_1 is $1/\{n_1\pi_1(1-\pi_1)\} + 1/\{n_2\pi_2(1-\pi_2)\}$. For the fit of Example 7.7, with $\bar{y}_1$ and $\bar{y}_2$ in place of π_1 and π_2:

```r
n_m <- sum(tab_sex[, "0"]); n_w <- sum(tab_sex[, "1"]); round(c(se_formula = sqrt(1 / (n_m * p_m * (1 - p_m)) + 1 / (n_w * p_w * (1 - p_w))), se_glm = coef(summary(g1))[2, 2]), 6)
```

```
## se_formula     se_glm
##     0.1837     0.1837
```

The formula and **glm** give the same standard error, 0.1837. ■

Exercise 7.5 (FREES exercise 11.5)**.** With $\hat{y}_i = \pi(\mathbf{x}_i'\mathbf{b})$ the logit fitted values of a model with an intercept, show that $\bar{y} = n^{-1}\sum_i \hat{y}_i$.

Solution. The score equation for the intercept, Corollary 7.16, is $\sum_i \hat{y}_i = \sum_i y_i$. For the reduced model of Example 7.13:

```r
c(mean_response = mean(H$POSEXP), mean_fitted = mean(fitted(red)))
```

```
## mean_response   mean_fitted
##        0.0785        0.0785
```

■

Exercise 7.6 (FREES exercise 11.6)**.** Beginning with the score equations (11.4) of FREES, verify the logit form (11.5).

Solution. Equation (11.4) is the score of Theorem 7.14, whose summands carry the factor $\pi'(\eta_i)/\{\pi_i(1-\pi_i)\}$. For the logit $\pi' = \Lambda' = \Lambda(1-\Lambda)$ by Proposition 7.8, so the factor is 1 and the score is $\sum_i \mathbf{x}_i(y_i - \pi_i)$, equation (11.5). ■

Exercise 7.7 (FREES exercise 11.7)**.** (a) From the logit score, show that $\mathcal{I}(\boldsymbol{\beta}) = \sum_i \sigma_i^2 \mathbf{x}_i \mathbf{x}_i'$ with $\sigma_i^2 = \pi_i(1-\pi_i)$. (b) From the general score (11.4), find the information matrix.

Solution. (a) Differentiating $\sum_i \mathbf{x}_i(y_i - \pi_i)$, with $\partial\pi_i/\partial\boldsymbol{\beta}' = \pi_i(1-\pi_i)\mathbf{x}_i'$, gives $\nabla^2 L = -\sum_i \pi_i(1-\pi_i)\mathbf{x}_i\mathbf{x}_i'$, which contains no response and is minus its own expectation. (b) With $r = \pi'/\{\pi(1-\pi)\}$ the score is $\sum_i \mathbf{x}_i(y_i - \pi_i)r(\eta_i)$, and

$$\nabla^2 L = \sum_i \mathbf{x}_i\mathbf{x}_i'\{-\pi'(\eta_i)r(\eta_i) + (y_i - \pi_i)r'(\eta_i)\};$$

the second term has mean zero, so $\mathcal{I}(\boldsymbol{\beta}) = -\mathsf{E}\,\nabla^2 L = \sum_i \pi'(\eta_i)^2 \mathbf{x}_i\mathbf{x}_i'/\{\pi_i(1-\pi_i)\}$, which is Theorem 7.14. ■

Exercise 7.8 (FREES exercise 11.8)**.** The IRC's $n = 1340$ bodily injury claims; `ATTORNEY` is 1 if the claimant is represented by an attorney and 2 if not. The original file was withdrawn; `AUTOBIsim`, its simulated substitute, is used. (a)(i) Histograms and summary statistics of `CLMAGE` and `LOSS`, and a variable `lnLOSS`; (ii) their means by `ATTORNEY`; (iii) percentages of `ATTORNEY` by `CLMSEX`, `MARITAL`, `SEATBELT` and `CLMINSUR`; (iv) missing values. (b)(i) A logit model on `CLMSEX`, interpreted through odds; (ii) a logit model on `CLMAGE`, `CLMSEX`, `MARITAL`, `SEATBELT`, `CLMINSUR`: which variables are significant? (iii) who uses attorneys more, men or women? (iv) add `LOSS` and `lnLOSS`, keep the more important, and test it; (v) drop `CLMAGE` and compare, including the number of observations; (vi) the probability of an attorney for a single male aged 32 wearing a seat belt, the driver insured, with economic loss \$5,000, under the model of (iv). (c) Part (b)(v) by probit, interpreting only signs.

Solution.

```
bi <- read.csv("data/frees/csv/AUTOBIsim.csv"); bi$lnLOSS <- log(bi$LOSS); bi$att <-
 ↪  1 * (bi$ATTORNEY == 1)
round(sapply(bi[, c("CLMAGE", "LOSS", "lnLOSS")], function(v) c(summary(v), sd =
 ↪  sd(v))), 3)
```

```
##          CLMAGE    LOSS lnLOSS
## Min.      0.100   0.010 -4.605
## 1st Qu.   1.800   0.630 -0.462
## Median    3.350   2.000  0.693
## Mean      5.801   5.682  0.603
## 3rd Qu.   7.000   5.438  1.693
## Max.    161.100 341.720  5.834
## sd        8.822  16.174  1.547
```

```
round(sapply(bi[, c("CLMAGE", "LOSS", "lnLOSS")], function(v) tapply(v, bi$ATTORNEY,
 ↪  mean)), 3)
```

```
##   CLMAGE  LOSS lnLOSS
## 1  5.354 8.468  1.309
## 2  6.266 2.786 -0.131
```

```
lapply(bi[, c("CLMSEX", "MARITAL", "SEATBELT", "CLMINSUR")], function(v)
  round(100 * prop.table(table(ATTORNEY = bi$ATTORNEY, level = v), 2), 1))
```

```
## $CLMSEX
##          level
## ATTORNEY    1    2
```

```
##        1 59.4 44.9
##        2 40.6 55.1
##
## $MARITAL
##          level
## ATTORNEY    1    2    3    4
##        1 51.6 49.4 35.7 73.5
##        2 48.4 50.6 64.3 26.5
##
## $SEATBELT
##          level
## ATTORNEY    1    2
##        1 50.9 54.5
##        2 49.1 45.5
##
## $CLMINSUR
##          level
## ATTORNEY    1    2
##        1 74.1 48.9
##        2 25.9 51.1
```

```r
colSums(is.na(bi))
```

```
##         X ATTORNEY    CLMSEX   MARITAL CLMINSUR SEATBELT
##         0        0         0         0        0        0
##   CLMAGE     LOSS   lnLOSS       att
##         0        0         0         0
```

```r
op <- par(mfrow = c(1, 3), mar = c(3.2, 3.4, 1.6, 0.6), mgp = c(2, 0.6, 0))
for (v in c("CLMAGE", "LOSS", "lnLOSS")) hist(bi[[v]], breaks = 40, main = v, xlab =
 ↪  "", col = "grey85", border = "white")
par(op)
```

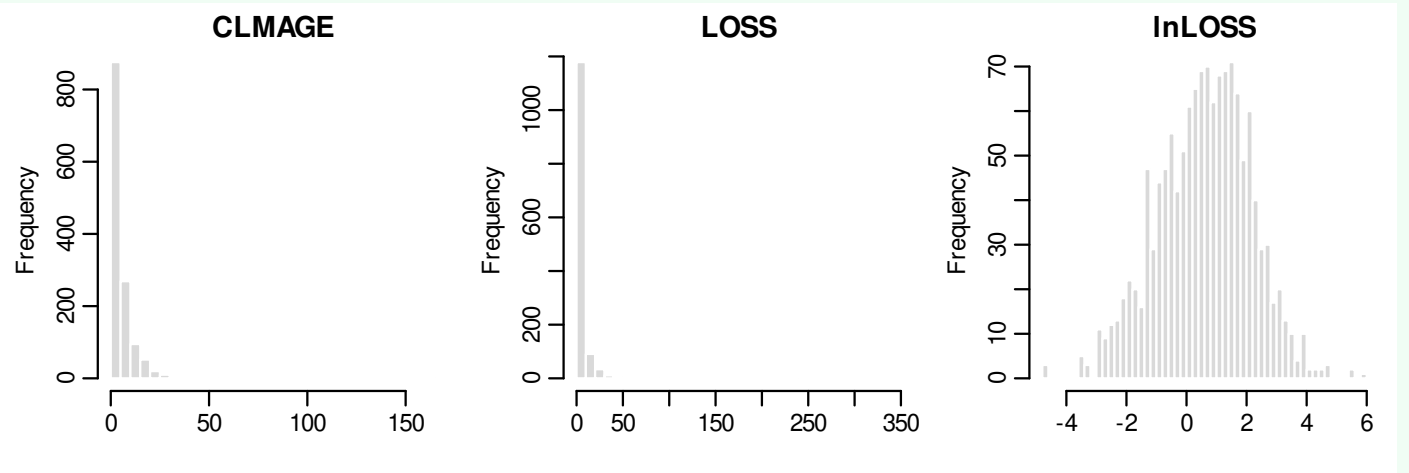


Figure 7.6: Histograms of `CLMAGE`, `LOSS` and `lnLOSS` in `AUTOBIsim`.

In the substitute file `CLMAGE` is not an age: its median is 3.350 and its maximum 161.100, and the questions about age are answered for this variable as it stands. `LOSS` is skewed (mean 5.682, median 2.000, maximum 341.720) and `lnLOSS` nearly symmetric (mean 0.603, median 0.693), so `lnLOSS` is the useful scale. Claimants with an attorney have larger losses (mean `lnLOSS` 1.309 against -0.131) and slightly smaller `CLMAGE` (5.354 against 6.266). Attorneys are used by 59.4%

of men and 44.9% of women, 73.5% of the divorced or separated, and 74.1% of claimants whose driver was uninsured against 48.9% of the others, while seat belts make little difference (50.9% and 54.5%). The substitute has no missing values.

```
for (v in c("CLMSEX", "MARITAL", "SEATBELT", "CLMINSUR")) bi[[v]] <- factor(bi[[v]])
m1 <- glm(att ~ CLMSEX, family = binomial, data = bi); round(coef(summary(m1)), 4)
```

```
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept)   0.3818     0.0859   4.444        0
## CLMSEX2      -0.5882     0.1121  -5.245        0
```

```
c(odds_ratio_female = exp(coef(m1)[["CLMSEX2"]]))
```

```
## odds_ratio_female
##            0.5553
```

```
m2 <- glm(att ~ CLMAGE + CLMSEX + MARITAL + SEATBELT + CLMINSUR, family = binomial,
 ↪  data = bi); round(coef(summary(m2)), 4)
```

```
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept)   1.4858     0.2406  6.1756   0.0000
## CLMAGE       -0.0101     0.0070 -1.4282   0.1532
```

```
## CLMSEX2      -0.5641     0.1146 -4.9215   0.0000
## MARITAL2     -0.1319     0.1143 -1.1542   0.2484
## MARITAL3     -0.3258     0.5661 -0.5755   0.5649
## MARITAL4      1.0186     0.4003  2.5448   0.0109
## SEATBELT2     0.0622     0.4395  0.1414   0.8875
## CLMINSUR2    -1.1044     0.2259 -4.8879   0.0000
```

```
c(odds_ratio_female_model_ii = exp(coef(m2)[["CLMSEX2"]]))
```

```
## odds_ratio_female_model_ii
##                     0.5689
```

```
m4 <- glm(att ~ CLMAGE + CLMSEX + MARITAL + SEATBELT + CLMINSUR + LOSS + lnLOSS,
 ↪  family = binomial, data = bi)
round(coef(summary(m4))[c("LOSS", "lnLOSS"), ], 4)
```

```
##         Estimate Std. Error z value Pr(>|z|)
## LOSS     -0.0114     0.0047  -2.425   0.0153
## lnLOSS    0.8520     0.0585  14.564   0.0000
```

```
m4b <- glm(att ~ CLMAGE + CLMSEX + MARITAL + SEATBELT + CLMINSUR + lnLOSS, family =
 ↪  binomial, data = bi)
```

```
round(coef(summary(m4b)), 4)
```

```
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)    0.7818     0.2707   2.888   0.0039
## CLMAGE        -0.0087     0.0069  -1.260   0.2077
## CLMSEX2       -0.7035     0.1310  -5.372   0.0000
## MARITAL2       0.1849     0.1314   1.408   0.1592
## MARITAL3      -0.7737     0.6982  -1.108   0.2678
## MARITAL4       0.6414     0.4461   1.438   0.1505
## SEATBELT2     -0.9964     0.4896  -2.035   0.0418
## CLMINSUR2     -0.9202     0.2520  -3.652   0.0003
## lnLOSS         0.7984     0.0527  15.156   0.0000
```

```
m5 <- update(m4b, . ~ . - CLMAGE)
round(coef(summary(m5)), 4)
```

```
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)    0.7359     0.2672   2.754   0.0059
## CLMSEX2       -0.7101     0.1307  -5.431   0.0000
## MARITAL2       0.1758     0.1310   1.342   0.1797
## MARITAL3      -0.8088     0.6960  -1.162   0.2452
```

```
## MARITAL4      0.6629     0.4452   1.489   0.1365
## SEATBELT2    -0.9739     0.4897  -1.989   0.0467
## CLMINSUR2    -0.9181     0.2509  -3.660   0.0003
## lnLOSS        0.7984     0.0526  15.171   0.0000
```

```
c(n_model_iv = nobs(m4b), n_model_v = nobs(m5))
```

```
## n_model_iv  n_model_v
##       1340       1340
```

```
new8 <- data.frame(CLMAGE = 32, CLMSEX = factor(1, levels = 1:2), MARITAL =
 ↪  factor(2, levels = 1:4),
                   SEATBELT = factor(1, levels = 1:2), CLMINSUR = factor(2, levels =
                    ↪  1:2), lnLOSS = log(5))
c(probability_attorney = unname(predict(m4b, new8, type = "response")),
 ↪  share_CLMAGE_below_32 = mean(bi$CLMAGE < 32))
```

```
##  probability_attorney share_CLMAGE_below_32
##                0.7412                0.9881
```

```
m5p <- glm(formula(m5), family = binomial(link = "probit"), data = bi);
 ↪  round(cbind(logit = coef(m5), probit = coef(m5p)), 4)
```

```
##                logit  probit
## (Intercept)  0.7359  0.4388
## CLMSEX2     -0.7101 -0.4276
## MARITAL2     0.1758  0.1081
## MARITAL3    -0.8088 -0.5189
## MARITAL4     0.6629  0.4154
## SEATBELT2   -0.9739 -0.5687
## CLMINSUR2   -0.9181 -0.5528
## lnLOSS       0.7984  0.4760
```

(i) Sex is significant ($z = -5.245$): women's odds of using an attorney are 0.5553 times men's. (ii) `CLMSEX`, `CLMINSUR` and the divorced or separated level of `MARITAL` are significant at 5% (p-values below 0.0001, below 0.0001 and 0.0109); `CLMAGE` (0.1532), `SEATBELT` (0.8875) and the other marital levels are not. (iii) Men: the women's coefficient is -0.5641, so by Theorem 7.11(i) women's odds of using an attorney are 0.5689 times men's, the other variables held fixed. (iv) `lnLOSS` has $z = 14.564$ against -2.425 for `LOSS` and is kept; in the refit its z is 15.156. (v) Without `CLMAGE` the estimates move little (`CLMSEX` -0.7101 against -0.7035, `lnLOSS` 0.7984 in both), and both fits use all 1340 claims, the substitute having no missing values; in the original file

`CLMAGE` is missing for some claims, and dropping it enlarges the sample. (vi) 0.7412, at `CLMAGE` $= 32$ on the substitute's scale, a value above the fraction 0.9881 of its `CLMAGE` values: a prediction in the far tail. (c) Every probit coefficient has the sign of the logit one: women, claimants not wearing a seat belt and claimants whose driver was insured are less likely to use an attorney, and larger losses make it more likely. ■

Exercise 7.9 (FREES exercise 11.9)**.** Hong Kong horse racing: 925 races, one horse drawn at random from the ten in each; `FINISH` $= 1$ if it won, and `WIN` is the crowd's probability of its winning, read from the tote board. (a) A colleague would take two horses from each race: how are their responses related, and which assumption does this violate? (b) The mean of `FINISH`, the summary statistics of `WIN`, and the relation between the mean and the standard deviation of `FINISH`. (c) `WIN` by `FINISH`. (d) A linear probability model of `FINISH` on `WIN`: (i) significance; (ii) fit; (iii) can fitted values leave $[0, 1]$? (e) A logistic model: is `WIN` significant? (f) Fitted values of both models at `WIN` $= 0, 0.01, 0.05, 0.10, 1$, and a plot of one against the other. (g) Plot $\widehat{\texttt{FINISH}} - \texttt{WIN}$ against `WIN` and discuss a betting strategy.

Solution. (a) At most one horse wins a race, so the two responses have $y_1 y_2 = 0$ and $\text{cov}(y_1, y_2) = \mathsf{E}[y_1 y_2] - \pi_1 \pi_2 = -\pi_1 \pi_2 < 0$: they are dependent, against the independence of the responses (Definition 7.1).

```
hk <- read.csv("data/frees/csv/HKHorse.csv"); yb <- mean(hk$FINISH); nh <- nrow(hk)
round(c(mean_FINISH = yb, sd_FINISH = sd(hk$FINISH),
        formula = sqrt(nh * yb * (1 - yb) / (nh - 1)), mean_WIN = mean(hk$WIN),
        sd_WIN = sd(hk$WIN), min_WIN = min(hk$WIN), max_WIN = max(hk$WIN)), 4)
```

```
## mean_FINISH   sd_FINISH     formula    mean_WIN      sd_WIN
##      0.0951      0.2936      0.2936      0.0996      0.0819
##     min_WIN     max_WIN
##      0.0027      0.6500
```

```
round(sapply(split(hk$WIN, hk$FINISH), function(v) c(n = length(v), mean = mean(v),
 ↪  sd = sd(v))), 4)
```

```
##              0       1
## n    837.0000 88.0000
## mean   0.0931  0.1612
## sd     0.0764  0.1044
```

```r
lpm9 <- lm(FINISH ~ WIN, data = hk); lg9 <- glm(FINISH ~ WIN, family = binomial,
    data = hk)
round(rbind(lpm = coef(summary(lpm9))[2, ], logit = coef(summary(lg9))[2, ]), 4)
```

```
##       Estimate Std. Error t value Pr(>|t|)
## lpm     0.8749     0.1144   7.647        0
## logit   7.6467     1.1470   6.667        0
```

```r
round(c(R2_lpm = summary(lpm9)$r.squared, intercept_lpm = coef(lpm9)[[1]], at_WIN_1
    = sum(coef(lpm9))), 4)
```

```
##       R2_lpm intercept_lpm      at_WIN_1
##       0.0596        0.0080        0.8830
```

```r
wv <- c(0, 0.01, 0.05, 0.10, 1)
fl <- predict(lpm9, data.frame(WIN = wv))
fg <- predict(lg9, data.frame(WIN = wv), type = "response")
round(rbind(WIN = wv, lpm = fl, logit = fg, logit_over_WIN = fg / wv), 4)
```

```
##                     1      2      3      4      5
## WIN            0.0000 0.0100 0.0500 0.1000 1.0000
## lpm            0.0080 0.0168 0.0518 0.0955 0.8830
```

```
## logit          0.0398 0.0428 0.0572 0.0817 0.9886
## logit_over_WIN    Inf 4.2796 1.1447 0.8171 0.9886
```

```
wg <- seq(0.0005, 0.9995, by = 0.0005); up <- plogis(coef(lg9)[[1]] + coef(lg9)[[2]]
 ↪  * wg) > 1.25 * wg
ru <- rle(up); last <- cumsum(ru$lengths); first <- last - ru$lengths + 1
reg <- cbind(from = wg[first[ru$values]], to = wg[last[ru$values]])
cbind(reg, t(apply(reg, 1, function(ab) { k <- hk$WIN >= ab[1] & hk$WIN <= ab[2]
  c(horses = sum(k), won = sum(hk$FINISH[k]), expected_by_crowd =
   ↪  round(sum(hk$WIN[k]), 2)) })))
```

```
##         from     to horses won expected_by_crowd
## [1,] 0.0005 0.0435    274   6              6.75
## [2,] 0.4480 0.7360      4   1              2.04
```

```
op <- par(mfrow = c(1, 2), mar = c(3.2, 3.8, 0.6, 0.6), mgp = c(2.1, 0.6, 0))
plot(fitted(lg9), fitted(lpm9), pch = 16, cex = 0.4, xlab = "logit fitted", ylab =
 ↪  "LPM fitted"); abline(0, 1, lty = 3)
plot(hk$WIN, fitted(lg9) - hk$WIN, pch = 16, cex = 0.4, xlab = "WIN",
     ylab = "fitted - WIN"); abline(h = 0, lty = 3); par(op)
```

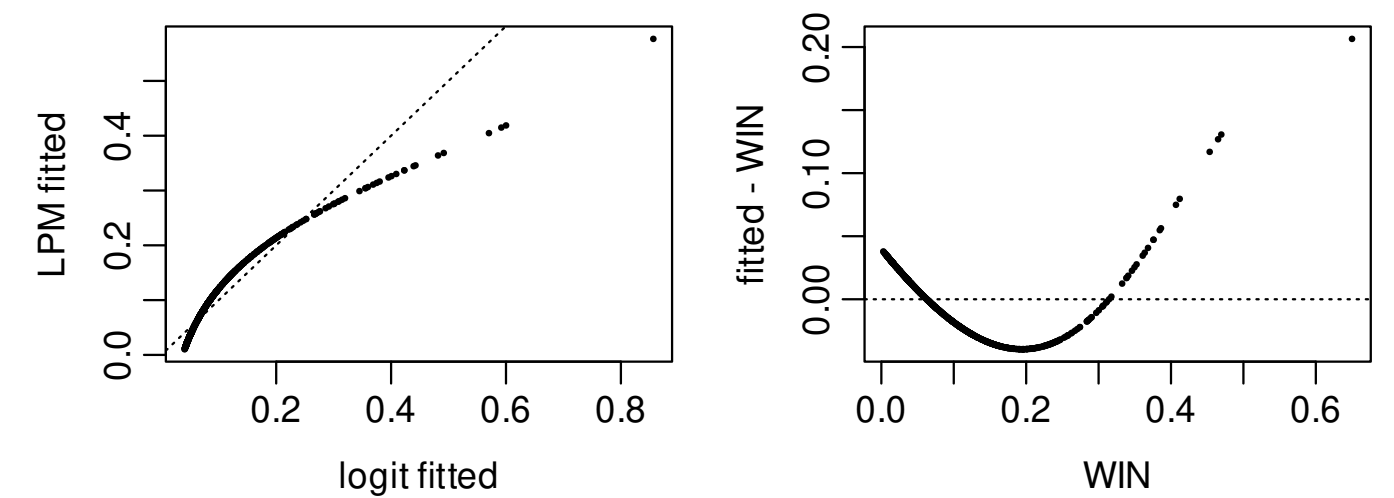


Figure 7.7: Left: fitted values of the linear probability model against those of the logit model, with the identity dotted. Right: the logit fitted probability minus the crowd's probability, against the crowd's probability.

(b) The mean of `FINISH` is 0.0951, near one in ten. For 0/1 data $\sum_i (y_i - \bar{y})^2 = \sum_i y_i - n\bar{y}^2 = n\bar{y}(1-\bar{y})$, so the standard deviation is $\{n\bar{y}(1-\bar{y})/(n-1)\}^{1/2} = 0.2936$, larger than that of `WIN`,

0.0819, though the means (0.0996 for `WIN`) are close. (c) Winners had mean `WIN` 0.1612 against 0.0931: the crowd's probabilities carry information. (d) `WIN` has $t = 7.647$, but R^2 is 0.0596; the fitted line is 0.0080 at `WIN` $= 0$ and 0.8830 at 1, so on $[0, 1]$ it stays in $[0, 1]$. (e) In the logit model `WIN` has $z = 6.667$. (f) The fits differ most at the ends: 0.0080 against 0.0398 at `WIN` $= 0$ and 0.8830 against 0.9886 at `WIN` $= 1$. (g) The winners share a fraction $1 - T$ of the pool in proportion to their stakes, so a unit bet on a horse carrying the fraction x of the bets returns $(1-T)/x$ if it wins, and its expected gain is positive iff the winning probability exceeds $x/(1-T)$, $1.25x$ for $T = 0.2$. On a grid of `WIN` the fitted logit probability exceeds 1.25 `WIN` on two intervals, from 0.0005 to 0.0435 and from 0.4480 to 0.7360 (the ratio is 4.2796 at 0.01, and 1.1447, 0.8171 and 0.9886 at 0.05, 0.10 and 1): the model suggests betting on extreme long shots and on strong favourites. The data do not bear this out: the 274 horses in the first interval won 6 races, against 6.75 expected from the crowd's probabilities, and the 4 in the second won 1, against 2.04, so the suggestion rests on the logit's functional form rather than on excess wins. ■

Exercise 7.10 (FREES exercise 11.10)**.** Term life insurance: the $n = 500$ SCF families with positive income, `FACEPOS` $= 1$ for the 275 that bought term life insurance. (a) Means of the

explanatory variables by FACEPOS. (b) A linear probability model on LINCOME, EDUCATION, AGE, GENDER and the factor MARSTAT: (i) define it; (ii) comment on its quality; (iii) its three drawbacks. (c) A logistic model on the same variables: (i) which variables are significant, and why; (ii) which measure summarises the fit? (d) A logistic model on LINCOME, EDUCATION and MARSTAT1, the indicator of MARSTAT $= 1$: (i) compare the two by a likelihood ratio test; (ii) who is more likely to buy, married or not, in terms of odds; (iii) the probability of buying for a married man aged 54 with 13 years of education, wages of \$70,000 and a household of four.

Solution.

```
tl <- read.csv("data/frees/csv/TermLife.csv"); tl$FACEPOS <- 1 * (tl$FACE > 0)
tl$LINCOME <- log(tl$INCOME); tl$MARSTAT1 <- 1 * (tl$MARSTAT == 1)
round(sapply(split(tl[, c("LINCOME", "EDUCATION", "AGE", "GENDER", "NUMHH",
 ↪  "MARSTAT1")], tl$FACEPOS), colMeans), 3)
```

```
##                0      1
## LINCOME   10.650 11.149
## EDUCATION 13.498 14.524
## AGE       46.347 47.833
## GENDER     0.778  0.865
```

```
## NUMHH       2.760  2.960
## MARSTAT1    0.556  0.756
```

```
lpm10 <- lm(FACEPOS ~ LINCOME + EDUCATION + AGE + GENDER + factor(MARSTAT), data =
↪  tl); round(coef(summary(lpm10)), 4)
```

```
##                   Estimate Std. Error t value Pr(>|t|)
## (Intercept)        -0.1011     0.1920 -0.5268   0.5986
## LINCOME             0.0263     0.0180  1.4636   0.1440
## EDUCATION           0.0193     0.0081  2.3845   0.0175
## AGE                -0.0002     0.0016 -0.1075   0.9144
## GENDER             -0.0128     0.0806 -0.1585   0.8741
## factor(MARSTAT)1    0.1724     0.0709  2.4326   0.0153
## factor(MARSTAT)2   -0.0500     0.1082 -0.4626   0.6439
```

```
round(c(R2 = summary(lpm10)$r.squared, min_fitted = min(fitted(lpm10)), max_fitted =
↪  max(fitted(lpm10)),
        outside_01 = sum(fitted(lpm10) < 0 | fitted(lpm10) > 1)), 4)
```

```
##         R2 min_fitted max_fitted outside_01
##     0.0690     0.1058     0.8012     0.0000
```

```r
lg10 <- glm(FACEPOS ~ LINCOME + EDUCATION + AGE + GENDER + factor(MARSTAT), family = binomial, data = tl)
round(coef(summary(lg10)), 4)
```

```
##                  Estimate Std. Error z value Pr(>|z|)
## (Intercept)       -2.5877     0.8453 -3.0614   0.0022
## LINCOME            0.1146     0.0788  1.4539   0.1460
## EDUCATION          0.0831     0.0354  2.3467   0.0189
## AGE               -0.0008     0.0070 -0.1205   0.9041
## GENDER            -0.0534     0.3437 -0.1553   0.8766
## factor(MARSTAT)1   0.7106     0.3011  2.3598   0.0183
## factor(MARSTAT)2  -0.2321     0.4756 -0.4881   0.6255
```

```r
L10 <- as.numeric(logLik(lg10)); L10_0 <- as.numeric(logLik(glm(FACEPOS ~ 1, family = binomial, data = tl)))
round(c(LRT_overall = 2 * (L10 - L10_0), pseudo_R2 = (L10 - L10_0) / (0 - L10_0), AIC = AIC(lg10)), 4)
```

```
## LRT_overall   pseudo_R2         AIC
##     35.2147      0.0512    666.9241
```

```r
lg10r <- glm(FACEPOS ~ LINCOME + EDUCATION + MARSTAT1, family = binomial, data =
  ↪  tl); round(coef(summary(lg10r)), 4)
```

```
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept)  -2.6964     0.7952  -3.391   0.0007
## LINCOME       0.1112     0.0781   1.424   0.1545
## EDUCATION     0.0860     0.0350   2.459   0.0139
## MARSTAT1      0.7207     0.2093   3.443   0.0006
```

```r
LR10 <- 2 * as.numeric(logLik(lg10) - logLik(lg10r))
round(c(LRT = LR10, df = 3, critical_5pct = qchisq(0.95, 3), p_value = pchisq(LR10,
  ↪  3, lower.tail = FALSE),
        odds_ratio_married = exp(coef(lg10r)[["MARSTAT1"]]),
        probability = unname(predict(lg10r, data.frame(LINCOME = log(70000),
         ↪  EDUCATION = 13, MARSTAT1 = 1), type = "response"))), 4)
```

```
##                LRT                 df       critical_5pct
##             0.4010             3.0000             7.8147
##            p_value odds_ratio_married        probability
##             0.9400             2.0558             0.5947
```

(a) Buyers have larger mean LINCOME (11.149 against 10.650) and EDUCATION (14.524 against

13.498), and a larger share married (0.756 against 0.556). (b)(i) The model of Definition 7.3, $\mathsf{P}(\texttt{FACEPOS} = 1) = \mathbf{x}'\boldsymbol{\beta}$, fitted by least squares. (ii) $R^2 = 0.0690$; only `EDUCATION` and married have $|t| > 2$; the fitted values stay within $[0.1058, 0.8012]$. (iii) The three drawbacks of Proposition 7.4. (c)(i) By the Wald test of Corollary 7.21, `EDUCATION` ($z = 2.3467$) and married (2.3598) are significant at 5%; `LINCOME` (1.4539), `AGE`, `GENDER` and living with a partner are not. (ii) The overall likelihood ratio statistic 35.2147, or the pseudo-R^2 0.0512 of Proposition 7.25; the AIC is 666.9241. (d)(i) H_0: the coefficients of `AGE`, `GENDER` and living with a partner are zero, the reduced model, against H_a: at least one of them is not, the full model; reject when LRT exceeds 7.8147, the 95% point of χ^2_3. LRT = 0.4010 ($p = 0.9400$), so H_0 is not rejected. (ii) Married persons: their odds of buying are 2.0558 times those of the others. (iii) 0.5947; age, sex and household size do not enter the reduced model. ■

Exercise 7.11 (FREES exercise 11.11)**.** Smith and Schumacher's logistic regression of graduating with an actuarial concentration for $n = 185$ students: intercept -12.094 (standard error 2.575), `GENDER` 0.256 (0.407), `PLACE%` 4.336 (1.657), `MSAT` 0.008 (0.004), `VSAT` -0.002 (0.003), `RANK` 4.442 (1.836). (a) Compute the t ratios and state which variables are significant. (b) With `GENDER` = 1,

compute the success probability for (i) PLACE% $= 0.80$, MSAT $= 680$, VSAT $= 570$, RANK $= 0.90$, and for the changes (ii) PLACE% $= 0.60$, (iii) MSAT $= 620$, (iv) VSAT $= 540$, (v) RANK $= 0.70$.

Solution.

```
est <- c(intercept = -12.094, GENDER = 0.256, PLACE = 4.336, MSAT = 0.008, VSAT =
 ↪  -0.002, RANK = 4.442)
se11 <- c(2.575, 0.407, 1.657, 0.004, 0.003, 1.836); round(c(est / se11, z_0.975 =
 ↪  qnorm(0.975)), 3)
```

```
## intercept    GENDER     PLACE      MSAT      VSAT      RANK
##    -4.697     0.629     2.617     2.000    -0.667     2.419
##   z_0.975
##     1.960
```

```
cases <- rbind(i   = c(1, 1, 0.80, 680, 570, 0.90),
               ii  = c(1, 1, 0.60, 680, 570, 0.90),
               iii = c(1, 1, 0.80, 620, 570, 0.90),
               iv  = c(1, 1, 0.80, 680, 540, 0.90),
               v   = c(1, 1, 0.80, 680, 570, 0.70))
round(plogis(drop(cases %*% est)), 4)
```

```
##      i     ii    iii     iv      v
## 0.4822 0.2812 0.3655 0.4972 0.2769
```

(a) By Corollary 7.21 with the normal reference, PLACE% (2.617), MSAT (2.000) and RANK (2.419) exceed 1.96 in size and are significant at 5%, MSAT barely; GENDER (0.629) and VSAT (−0.667) are not. (b) The probabilities are 0.4822 in case (i), 0.2812 with PLACE% 0.60, 0.3655 with MSAT 620, 0.4972 with VSAT 540 and 0.2769 with RANK 0.70. ■

Exercise 7.12 (FREES exercise 11.12)**.** Case-control sampling: from N records, each with $y_i = 1$ is kept with probability τ_1 and each with $y_i = 0$ with probability τ_0, the event $\{r_i = 1\}$. (a) Find $\mathsf{P}(y_i = 1, r_i = 1)$, $\mathsf{P}(y_i = 0, r_i = 1)$ and $\mathsf{P}(r_i = 1)$ in terms of τ_0, τ_1 and π_i. (b) Find $\mathsf{P}(y_i = 1 \mid r_i = 1)$. (c) Rewrite it for $\pi_i = \Lambda(\mathbf{x}_i'\boldsymbol{\beta})$. (d) Show that the likelihood of the kept responses is the logistic likelihood with a changed intercept, and give the intercept.

Solution. The selection depends on the record only through y_i: $\mathsf{P}(r_i = 1 \mid y_i = 1) = \tau_1$ and $\mathsf{P}(r_i = 1 \mid y_i = 0) = \tau_0$. (a) $\mathsf{P}(y_i = 1, r_i = 1) = \tau_1\pi_i$, $\mathsf{P}(y_i = 0, r_i = 1) = \tau_0(1-\pi_i)$, and their sum $\mathsf{P}(r_i = 1) = \tau_1\pi_i + \tau_0(1-\pi_i)$. (b) $\mathsf{P}(y_i = 1 \mid r_i = 1) = \tau_1\pi_i/\{\tau_1\pi_i + \tau_0(1-\pi_i)\}$. (c) Multiplying numerator and denominator by $1 + e^{\eta_i}$, $\eta_i = \mathbf{x}_i'\boldsymbol{\beta}$, gives $\tau_1 e^{\eta_i}/(\tau_1 e^{\eta_i} + \tau_0) = \Lambda\{\eta_i + \ln(\tau_1/\tau_0)\}$. (d)

Given the selection, the kept responses are independent with these probabilities, so their likelihood is that of Definition 7.13 with the logit and $\mathbf{x}_i'\boldsymbol{\beta}$ replaced by $(\beta_0 + \ln(\tau_1/\tau_0)) + \sum_{j\geqslant 1} x_{ij}\beta_j$: the slopes are unchanged and the intercept becomes $\beta_0 + \ln(\tau_1/\tau_0)$. A simulation with $\tau_1 = 1$ and $\tau_0 = 0.1$:

```
set.seed(1112); N <- 2000000; xx <- rnorm(N); yy <- rbinom(N, 1, plogis(-4 + xx))
keep <- runif(N) < ifelse(yy == 1, 1, 0.1)
round(c(intercept_shift = coef(glm(yy[keep] ~ xx[keep], family = binomial))[[1]] +
 ↪  4, log_10 = log(10),
        slope = coef(glm(yy[keep] ~ xx[keep], family = binomial))[[2]]), 3)
```

```
## intercept_shift          log_10           slope
##           2.287           2.303           1.010
```

The estimated intercept shift is 2.287, against $\ln 10 = 2.303$, and the slope 1.010, against 1. ■

Exercise 7.13 (SOA sample question 19)**.** For $Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_1 X_2 + \varepsilon$ the maximised log-likelihoods are -1126.91 with the intercept only, -1122.41 with the intercept, X_1 and X_2, and -1121.91 with all four terms. $H_0\colon \beta_1 = \beta_2 = \beta_3 = 0$ is tested at the 5% level by the likelihood ratio test. Which is true? (A) The statistic is 1 and H_0 cannot be rejected. (B) It is 9

and H_0 cannot be rejected. (C) It is 10 and H_0 cannot be rejected. (D) It is 9 and H_0 should be rejected. (E) It is 10 and H_0 should be rejected.

Solution. H_0 restricts the full model to the intercept-only model, $q = 3$, so the statistic is $2\{-1121.91 - (-1126.91)\}$, referred to χ^2_3 by Proposition 7.22:

```
round(c(LRT = 2 * (-1121.91 + 1126.91), critical_5pct = qchisq(0.95, 3), stat_1 = 2
 ↪  * (-1121.91 + 1122.41),
       stat_9 = 2 * (-1122.41 + 1126.91), p_value = pchisq(10, 3, lower.tail =
        ↪  FALSE)), 4)
```

```
##          LRT critical_5pct       stat_1       stat_9
##      10.0000       7.8147       1.0000       9.0000
##      p_value
##       0.0186
```

The statistic is 10.0000, above the 5% critical value 7.8147 of χ^2_3 (p-value 0.0186), so H_0 is rejected: (E). The 1 of (A) compares the full model with the model without the interaction and tests $\beta_3 = 0$ only; the 9 of (B) and (D) compares the model without the interaction with the intercept-only one and tests $\beta_1 = \beta_2 = 0$ given $\beta_3 = 0$; (C) has the right statistic and the wrong decision. ■

Exercise 7.14 (SOA sample question 42)**.** Which statement is NOT true about the linear probability, logistic and probit models for binary responses? (A) The three major drawbacks of the linear probability model are poor fitted values, heteroscedasticity and meaningless residual analysis. (B) The logistic and probit models aim to circumvent the drawbacks of linear probability models. (C) The logit function is $\pi(z) = e^z/(1+e^z)$. (D) The probit function is $\pi(z) = \Phi(z)$. (E) The logit and probit functions are substantially different.

Solution. (A) is true: Proposition 7.4. (B) is true: their fitted probabilities lie in $(0,1)$, and the likelihood of Definition 7.13 uses the Bernoulli law of each response, with its variance $\pi_i(1-\pi_i)$, instead of a least squares fit with constant variance and continuous errors. (C) and (D) are the definitions (Definition 7.6, Proposition 7.8). (E) is false: after rescaling by $\pi/\sqrt{3}$ the two distribution functions differ by at most 0.0227, and $\text{logit}(p)/\Phi^{-1}(p)$ stays between 1.5958 and 1.7145 on $[0.1, 0.9]$ (Example 7.6). The answer is (E). ■

Exercise 7.15 (SOA sample question 67)**.** A succeeds 8 times out of 10 and B 2 times out of 10. A logistic model with linear term $\beta_0 + \beta_1 X$, X the indicator of A, estimates success probabilities 0.8 for A and 0.2 for B; under $\beta_1 = 0$ both are 0.5. By the likelihood ratio test, what is the lowest

of the significance levels (A) 0.100, (B) 0.050, (C) 0.025, (D) 0.010, (E) 0.005 at which $\beta_1 = 0$ is rejected?

Solution. By Corollary 7.16 the fitted probabilities are the two proportions and, under $\beta_1 = 0$, the pooled 10/20, as stated. The full model gives probability 0.8 to $8 + 8 = 16$ outcomes and 0.2 to 4; the reduced model 0.5 to all 20. The statistic is referred to χ^2_1 (Corollary 7.21):

```
LR67 <- 2 * ((16 * log(0.8) + 4 * log(0.2)) - 20 * log(0.5))
round(c(LRT = LR67, qchisq(1 - c(`0.100` = 0.1, `0.050` = 0.05, `0.025` = 0.025,
 ↪  `0.010` = 0.01, `0.005` = 0.005), 1)), 3)
```

```
##   LRT 0.100 0.050 0.025 0.010 0.005
## 7.710 2.706 3.841 5.024 6.635 7.879
```

The statistic 7.710 exceeds the critical values of χ^2_1 at 0.100, 0.050, 0.025 and 0.010 (2.706, 3.841, 5.024, 6.635) but not the 7.879 at 0.005, so the lowest available level at which H_0 is rejected is 0.010: (D). ■

Exercise 7.16 (SOA sample question 72)**.** A logistic model of positive medical expenditure gives: Age 0.257 (standard error 0.102, p-value 0.0117); Sex −0.955 (0.542, 0.0781); Education 0.103

(0.069, 0.1355); Log(Income) −0.516 (0.157, 0.0010). Which predictors are significant at 5%? (A) Log(Income) only; (B) Education only; (C) Age and Log(Income) only; (D) Sex and Education only; (E) all four.

Solution. Each p-value is $2\{1-\Phi(|z|)\}$ with $z = b_j/\operatorname{se}(b_j)$, the Wald test of Corollary 7.21 with $q = 1$:

```
b72 <- c(Age = 0.257, Sex = -0.955, Education = 0.103, LogIncome = -0.516); s72 <-
 ↪  c(0.102, 0.542, 0.069, 0.157)
round(rbind(z = b72 / s72, p_value = 2 * pnorm(-abs(b72 / s72))), 4)
```

```
##             Age     Sex Education LogIncome
## z        2.5196 -1.7620    1.4928    -3.287
## p_value 0.0117  0.0781    0.1355     0.001
```

The z ratios 2.5196, −1.7620, 1.4928 and −3.287 give back the p-values 0.0117, 0.0781, 0.1355 and 0.001. Age and Log(Income) are below 0.05 and Sex and Education are not, so (C) holds, and (A), (B), (D) and (E) each misclassify at least one predictor. ■

Exercise 7.17 (ISLR §4.8, exercise 1)**.** Show that the logistic function representation $p(X) = e^{\beta_0+\beta_1 X}/(1+e^{\beta_0+\beta_1 X})$ and the logit representation $p(X)/\{1-p(X)\} = e^{\beta_0+\beta_1 X}$ are equivalent.

Solution. Write $z = \beta_0 + \beta_1 X$. If $p = e^z/(1+e^z)$ then $1-p = 1/(1+e^z)$ and $p/(1-p) = e^z$. If $p/(1-p) = e^z$ with $p \in (0,1)$, then $p = e^z(1-p)$, so $p(1+e^z) = e^z$ and $p = e^z/(1+e^z)$. This is logit $\Lambda(z) = z$ of Proposition 7.8. ■

Exercise 7.18 (ISLR §4.8, exercise 3)**.** With one predictor and K classes, let X be normal in class k with mean μ_k and variance σ_k^2, the variances differing between classes, and let π_k be the proportion of class k. Show that the Bayes classifier is not linear in x but quadratic.

Solution. Write π_k for the class proportions and f_k for the density of X in class k, with $f = \sum_l \pi_l f_l > 0$. **Bayes' theorem** in this form, $\mathsf{P}(Y = k \mid X = x) = \pi_k f_k(x)/f(x)$ (ISLR (4.15)), holds because for every bounded g, $\mathsf{E}[g(X)I(Y=k)] = \pi_k \int g f_k = \mathsf{E}[g(X)\,\pi_k f_k(X)/f(X)]$, so $\pi_k f_k(X)/f(X)$ satisfies the defining relation of the conditional expectation of $I(Y=k)$ given X (the prerequisites' lemma **conditional expectation in** L^1); with p_k the Borel function of Unit 1's convention, the uniqueness argument of that lemma needs the relation only for $W = I\{h(X) >$

$p_k(X)\}$ and its mirror, $h = \pi_k f_k / f$, which are bounded functions of X. Here f_k is the normal density with mean μ_k and variance σ_k^2, and the denominator $f(x)$ is the same for every k, so the Bayes classifier (Unit 1) assigns x to the k maximising $\pi_k f_k(x)$, or its logarithm, which apart from a constant common to all classes is

$$\delta_k(x) = \ln \pi_k - \ln \sigma_k - \frac{(x-\mu_k)^2}{2\sigma_k^2} = -\frac{x^2}{2\sigma_k^2} + \frac{\mu_k}{\sigma_k^2} x - \frac{\mu_k^2}{2\sigma_k^2} + \ln \pi_k - \ln \sigma_k .$$

The coefficient of x^2 depends on k, so the boundary $\delta_k(x) = \delta_l(x)$ between two classes is the solution of a quadratic equation in x; only when $\sigma_k = \sigma_l$ do the x^2 terms cancel and the boundary become linear. ■

Exercise 7.19 (ISLR §4.8, exercise 4)**.** Let X be uniform on $[0,1]^p$, and predict at a test point from the training observations within 10% of the range of each feature closest to it. (a) For $p = 1$, what fraction of the observations is used on average? (b) For $p = 2$? (c) For $p = 100$? (d) What does this say about KNN for large p? (e) For a hypercube centred at the test point containing on average 10% of the observations, find the side length for $p = 1$, 2, 100.

Solution. (a) With the window $[x_0 - 0.05, x_0 + 0.05]$ clipped to $[0, 1]$, its length is 0.1 for $x_0 \in [0.05, 0.95]$ and $0.05 + x_0$ or $1.05 - x_0$ near the ends, so the average fraction is $0.1 - 2\int_0^{0.05}(0.05 - t)\,dt = 0.1 - 0.0025 = 0.0975$; with the window shifted to stay inside $[0, 1]$, as ISLR's "closest 10% of the range" reads, it is 0.1 exactly. (b) The features are independent, so the fraction is the product: $0.0975^2 = 0.0095$, or 0.01. (c) $0.0975^{100} = 8 \times 10^{-102}$, or 10^{-100}. (d) For large p a neighbourhood that is local in every coordinate contains essentially no training points, so the K nearest neighbours are far away, Unit 6's proposition **neighbourhoods are not local**. (e) Side s with $s^p = 0.1$: $s = 0.1$, 0.316 and $0.1^{1/100} = 0.977$; at $p = 100$ the cube that holds 10% of the data spans almost the whole range of every feature. ■

Exercise 7.20 (ISLR §4.8, exercise 6)**.** A logistic regression of receiving an A on X_1 = hours studied and X_2 = undergraduate GPA gives $\hat\beta_0 = -6$, $\hat\beta_1 = 0.05$, $\hat\beta_2 = 1$. (a) Estimate the probability of an A for a student who studies 40 hours with GPA 3.5. (b) How many hours does that student need for a 50% chance?

Solution. (a) $\Lambda(-6 + 0.05 \cdot 40 + 3.5) = \Lambda(-0.5) = 1/(1 + e^{0.5}) = 0.3775$. (b) $\Lambda(z) = 1/2$ iff $z = 0$: $-6 + 0.05h + 3.5 = 0$, $h = 50$ hours. ■

Exercise 7.21 (ISLR §4.8, exercise 7)**.** Companies are to be classified by whether they issue a dividend (Y = Yes or No) from last year's percentage profit X. 80% issued dividends; X is normal with mean 10 for those that did and 0 for those that did not, with variance 36 in both. Predict the probability that a company with $X = 4$ issues a dividend.

Solution. By Bayes' theorem in the form of Exercise 7.18's solution, with ϕ the normal density with standard deviation 6,

$$\mathsf{P}(Y = \text{Yes} \mid X = 4) = \frac{0.8\,\phi(4-10)}{0.8\,\phi(4-10) + 0.2\,\phi(4-0)} = \frac{0.8e^{-1/2}}{0.8e^{-1/2} + 0.2e^{-2/9}} = 0.752.$$

The class probability of Unit 1's Bayes classifier is computed here from the class densities and the prior proportions, the route of ISLR's §4.4, rather than modelled directly as in logistic regression. ■

Exercise 7.22 (ISLR §4.8, exercise 8)**.** A data set is split into equal training and test halves. Logistic regression has training error 20% and test error 30%; 1-nearest neighbours has error 18% averaged over the two halves. Which method should be preferred for new observations?

Solution. With distinct predictor values, each training point is its own nearest neighbour, so the 1-NN training error is 0 and the average 18% means a test error of 36%. Logistic regression's test error, 30%, is lower, and the test error estimates the error on new observations (Unit 1); prefer logistic regression. ■

Exercise 7.23 (ISLR §4.8, exercise 9)**.** (a) What fraction of people with odds 0.37 of defaulting default on average? (b) What are the odds for a person with a 16% chance of defaulting?

Solution. By Definition 7.7, odds o correspond to $p = o/(1+o)$. (a) $0.37/1.37 = 0.270$. (b) $0.16/0.84 = 0.190$. ■

Exercise 7.24 (ISLR §4.8, exercise 12)**.** You fit $\widehat{\mathsf{P}}(\texttt{orange} \mid x) = \Lambda(\hat{\beta}_0 + \hat{\beta}_1 x)$ and a friend fits the softmax coding with $\hat{\alpha}_{\texttt{orange0}}, \hat{\alpha}_{\texttt{orange1}}, \hat{\alpha}_{\texttt{apple0}}, \hat{\alpha}_{\texttt{apple1}}$. (a) What are the log odds of orange against apple in your model? (b) In your friend's? (c) If $\hat{\beta}_0 = 2$, $\hat{\beta}_1 = -1$, what are your friend's estimates? (d) If your friend's are 1.2, −2, 3, 0.6, what are yours? (e) On 2000 test observations, how often do the two models' predicted labels agree?

Solution. (a) $\hat\beta_0 + \hat\beta_1 x$. (b) $(\hat\alpha_{\texttt{orange0}} - \hat\alpha_{\texttt{apple0}}) + (\hat\alpha_{\texttt{orange1}} - \hat\alpha_{\texttt{apple1}})x$ (Proposition 7.30). (c) Only the differences are determined: $\hat\alpha_{\texttt{orange0}} - \hat\alpha_{\texttt{apple0}} = 2$ and $\hat\alpha_{\texttt{orange1}} - \hat\alpha_{\texttt{apple1}} = -1$; adding a common vector to both classes changes no probability (Proposition 7.28), so the four numbers are not identified. (d) $\hat\beta_0 = 1.2 - 3 = -1.8$ and $\hat\beta_1 = -2 - 0.6 = -2.6$ (Corollary 7.29). (e) The two fits give the same probability at every x, so the labels agree on all 2000 observations, except at a probability of exactly 1/2. ■

Exercise 7.25 (ISLR §4.8, exercise 13, parts (a)–(d), (g), (i), and (j) without LDA, QDA and naive Bayes)**.** With `Weekly` (1089 weekly returns, 1990–2010): (a) summarise the data; (b) fit the logit of `Direction` on the five lags and `Volume`, and identify any significant predictor; (c) give the confusion matrix and the fraction of correct predictions; (d) fit on 1990–2008 with `Lag2` only and evaluate on 2009–2010; (g) repeat (d) with KNN, $K = 1$; (i) which method is best? (j) Experiment with other predictors, transformations and interactions in the logit, and other values of K for KNN, reporting the test accuracy. Parts (e), (f), (h) and the rest of (j) use LDA, QDA and naive Bayes, which are on neither reading list and are not covered.

Solution.

```
suppressMessages(library(class)); Wk <- Weekly
round(c(n = nrow(Wk), cor_year_volume = cor(Wk$Year, Wk$Volume), up =
 ↪  mean(Wk$Direction == "Up")), 3)
```

```
##              n cor_year_volume             up
##       1089.000          0.842          0.556
```

```
gW <- glm(Direction ~ Lag1 + Lag2 + Lag3 + Lag4 + Lag5 + Volume, data = Wk, family =
 ↪  binomial)
round(coef(summary(gW)), 4)
```

```
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept)   0.2669     0.0859  3.1056   0.0019
## Lag1         -0.0413     0.0264 -1.5626   0.1181
## Lag2          0.0584     0.0269  2.1754   0.0296
## Lag3         -0.0161     0.0267 -0.6024   0.5469
## Lag4         -0.0278     0.0265 -1.0501   0.2937
## Lag5         -0.0145     0.0264 -0.5485   0.5833
## Volume       -0.0227     0.0369 -0.6163   0.5377
```

```
pW <- cls(fitted(gW)); table(pW, Wk$Direction); mean(pW == Wk$Direction)
```

```
##
## pW      Down  Up
##   Down    54  48
##   Up     430 557
## [1] 0.5611
```

```
trW <- Wk$Year <= 2008; dW <- Wk$Direction[!trW]
gW2 <- glm(Direction ~ Lag2, data = Wk, family = binomial, subset = trW)
pW2 <- cls(predict(gW2, Wk[!trW, ], type = "response")); table(pW2, dW)
```

```
##       dW
## pW2     Down Up
##   Down     9  5
##   Up      34 56
```

```
set.seed(1); kW <- knn(as.matrix(Wk$Lag2[trW]), as.matrix(Wk$Lag2[!trW]),
                       Wk$Direction[trW], k = 1); table(kW, dW)
```

```
##       dW
## kW      Down Up
##   Down    21 30
##   Up      22 31
```

```r
round(c(logit = mean(pW2 == dW), knn1 = mean(kW == dW), always_up = mean(dW ==
  "Up")), 4)
```

```
##     logit      knn1 always_up
##    0.6250    0.5000    0.5865
```

(a) `Volume` grows with `Year` (correlation 0.842); 55.6% of the weeks are `Up`. (b) Only `Lag2` is significant at 5% ($z = 2.18$, $p = 0.030$). (c) 56.1% of the weeks are classified correctly, but 987 of the 1089 are predicted `Up`: almost every `Down` week (430 of 484) is misclassified, while 557 of the 605 `Up` weeks are right. (d) On 2009–2010, 62.5% correct, against 58.7% for always predicting `Up`. (g) KNN with $K = 1$ is right 50.0% of the time. (i) The logit on `Lag2` is best of the two, and only modestly better than predicting `Up` every week.

```r
trw <- Wk$Year <= 2008
accw <- function(f) mean(ifelse(predict(glm(f, binomial, Wk[trw, ]), Wk[!trw, ],
  type = "response") > 0.5, "Up", "Down") == Wk$Direction[!trw])
round(c(lag2 = accw(Direction ~ Lag2),
        lag1_lag2 = accw(Direction ~ Lag1 + Lag2),
        lag2_squared = accw(Direction ~ Lag2 + I(Lag2^2)),
        lag2_times_volume = accw(Direction ~ Lag2 * Volume)), 4)
```

```
##              lag2         lag1_lag2      lag2_squared
##            0.6250            0.5769            0.6250
## lag2_times_volume
##            0.5385
```

```
set.seed(1)
round(sapply(c(K1 = 1, K5 = 5, K10 = 10, K20 = 20, K50 = 50), function(K)
  mean(knn(Wk[trw, "Lag2", drop = FALSE], Wk[!trw, "Lag2", drop = FALSE],
           Wk$Direction[trw], k = K) == Wk$Direction[!trw])), 4)
```

```
##     K1     K5    K10    K20    K50
## 0.5000 0.5385 0.5673 0.5962 0.5769
```

(j) Adding **Lag2**^{2} leaves the logit at 62.5%, and adding **Lag1** or the interaction with **Volume** lowers it; KNN improves with K up to about 20 (59.6%) but never reaches the logit on **Lag2**. On 104 test weeks each accuracy has standard error about $\sqrt{0.6 \cdot 0.4/104} = 0.048$ (Proposition 7.2, the weeks treated as independent), and a difference of two independent accuracies about $\sqrt{2}$ times that, 0.068 (differences of models scored on the same weeks have smaller standard errors, as low as 0.014 here, and the conclusion stands for them too); even the largest gap, 12.5 points between the logit and KNN with $K = 1$, is under two such standard errors, so none of the differences is

clearly more than noise, and the simplest model does as well as any. ■

Exercise 7.26 (ISLR §4.8, exercise 14, parts (a)–(c), (f), (h))**.** With `Auto`: (a) create `mpg01`, 1 when `mpg` exceeds its median; (b) find the features most associated with it; (c) split into training and test halves; (f) fit a logistic regression on those features and give its test error; (h) do the same with KNN for several K. Parts (d), (e), (g) use LDA, QDA and naive Bayes and are not covered.

Solution.

```
Au <- Auto; Au$mpg01 <- 1 * (Au$mpg > median(Au$mpg))
round(cor(Au[, c("mpg01", "cylinders", "displacement", "horsepower", "weight",
                 "acceleration", "year")])[1, -1], 3)
```

```
##    cylinders displacement   horsepower       weight
##       -0.759       -0.753       -0.667       -0.758
## acceleration         year
##        0.347        0.430
```

```
set.seed(1); teA <- sample(nrow(Au), nrow(Au) / 2)
gA <- glm(mpg01 ~ cylinders + displacement + horsepower + weight, data = Au[-teA, ],
          family = binomial)
```

```
XA <- scale(Au[, c("cylinders", "displacement", "horsepower", "weight")])
knn_err <- sapply(c(1, 3, 5, 10, 20, 50), function(K) { set.seed(1)
  mean(knn(XA[-teA, ], XA[teA, ], Au$mpg01[-teA], k = K) != Au$mpg01[teA]) })
round(c(logit = mean((predict(gA, Au[teA, ], type = "response") > 0.5) !=
  Au$mpg01[teA]),
        setNames(knn_err, paste0("knn", c(1, 3, 5, 10, 20, 50)))), 4)
```

```
##  logit   knn1   knn3   knn5  knn10  knn20  knn50
## 0.0969 0.0867 0.0765 0.0612 0.0765 0.0816 0.0816
```

(a), (b) **cylinders**, **displacement**, **weight** and **horsepower** have correlations -0.76 to -0.67 with **mpg01**, far larger than **acceleration** and **year**. (c) A random half of the 392 cars is the test set. (f) The logit's test error is 9.7%. (h) On the standardised features KNN's test error is 6.1% at $K = 5$, the best of the values tried, and 8.7% at $K = 1$. ■

Exercise 7.27 (ISLR §4.8, exercise 15)**.** Write functions: (a) **Power()** printing 2^3; (b) **Power2(x, a)** printing x^a; (c) use it for 10^3, 8^{17}, 131^3; (d) **Power3(x, a)** returning x^a; (e) plot $f(x) = x^2$ for $x = 1, \ldots, 10$, with a logarithmic axis; (f) **PlotPower(x, a)** plotting x against x^a.

Solution.

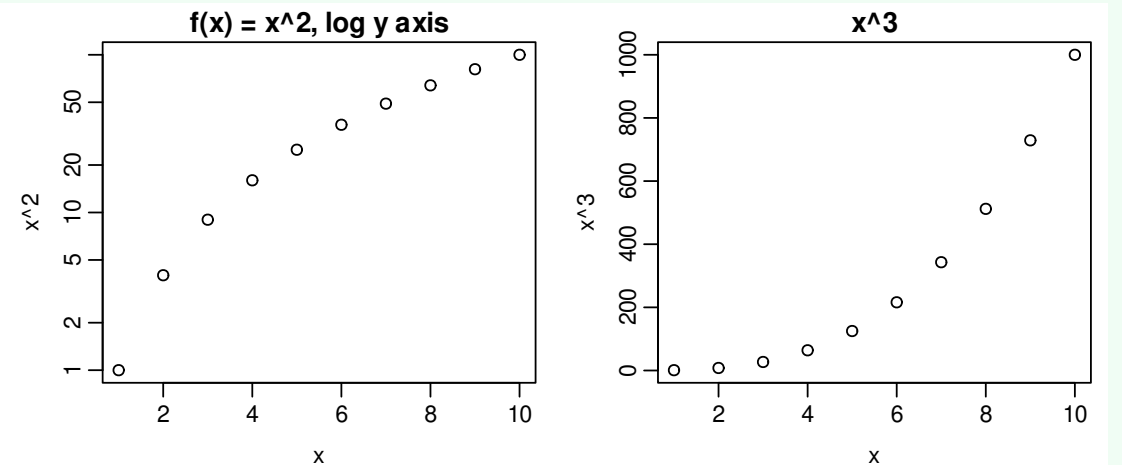
```
Power <- function() print(2^3); Power()
```

```
## [1] 8
```

```
Power2 <- function(x, a) print(x^a); Power2(3, 8); Power2(10, 3); Power2(8, 17);
 ↪  Power2(131, 3)
```

```
## [1] 6561
## [1] 1000
## [1] 2.252e+15
## [1] 2248091
```

```
Power3 <- function(x, a) { result <- x^a; return(result) }
PlotPower <- function(x, a, ...) plot(x, Power3(x, a), xlab = "x", ylab =
 ↪  paste0("x^", a), ...)
op <- par(mfrow = c(1, 2), mar = c(3.2, 3.8, 1.4, 0.6), mgp = c(2.1, 0.6, 0), cex =
 ↪  0.8)
PlotPower(1:10, 2, log = "y", main = "f(x) = x^2, log y axis")
PlotPower(1:10, 3, main = "x^3"); par(op)
```



`Power2` prints; `Power3` returns a value that can be used in further computation, which is what `PlotPower` needs. $8^{17} = 2.25 \times 10^{15}$ is printed in floating point. ■

Exercise 7.28 (ISLR §4.8, exercise 16, logistic regression and KNN)**.** With `Boston`, predict whether a tract's crime rate is above the median, by logistic regression and KNN on subsets of the predictors. (LDA and naive Bayes, also asked for, are not covered.)

Solution.

```
Bo <- Boston; Bo$crim01 <- 1 * (Bo$crim > median(Bo$crim))
round(cor(Bo)[setdiff(names(Bo), c("crim", "crim01")), "crim01"], 2)
```

```
##      zn   indus    chas     nox      rm     age     dis
##   -0.44    0.60    0.07    0.72   -0.16    0.61   -0.62
##     rad     tax ptratio   lstat    medv
##    0.62    0.61    0.25    0.45   -0.26
```

```
set.seed(1); teB <- sample(nrow(Bo), nrow(Bo) / 2)
err_logit <- function(f) { g <- glm(f, data = Bo[-teB, ], family = binomial)
  mean((predict(g, Bo[teB, ], type = "response") > 0.5) != Bo$crim01[teB]) }
XB <- scale(Bo[, c("nox", "rad", "dis", "age", "indus", "tax")])
knnB <- sapply(c(1, 3, 5, 10, 20), function(K) { set.seed(1)
  mean(knn(XB[-teB, ], XB[teB, ], Bo$crim01[-teB], k = K) != Bo$crim01[teB]) })
round(c(logit_all = err_logit(crim01 ~ . - crim),
        logit_six = err_logit(crim01 ~ nox + rad + dis + age + indus + tax),
        setNames(knnB, paste0("knn", c(1, 3, 5, 10, 20)))), 4)
```

```
## logit_all logit_six      knn1      knn3      knn5     knn10
##    0.1067    0.1383    0.0870    0.0751    0.0830    0.1067
```

```
##      knn20
##     0.1225
```

The predictors most correlated with **crim01** are **nox**, **rad**, **age**, **tax** and **indus** (positively) and **dis** (negatively). On a random test half, the logit on all predictors errs on 10.7% of the tracts and on the six correlated ones 13.8%; KNN on those six, standardised, does best at $K = 3$ with 7.5%, and worsens as K grows to 20 (12.3%). ■

Exercise 7.29 (ISLR §5.4, exercise 5)**.** On **Default**, (a) fit the logit of **default** on **income** and **balance**; (b) estimate its test error by the validation set approach; (c) repeat with three other splits; (d) add **student** and compare.

Solution. The validation set approach is Unit 1's; the classifier uses threshold 1/2:

```
vs <- t(sapply(1:4, function(sd) { set.seed(sd); tr <- sample(nrow(Df), nrow(Df) /
 ↪  2)
  e <- function(f) { g <- glm(f, data = Df, family = binomial, subset = tr)
    mean((predict(g, Df[-tr, ], type = "response") > 0.5) != Df$y[-tr]) }
  c(seed = sd, income_balance = e(default ~ income + balance),
    with_student = e(default ~ income + balance + student)) }))
```

```r
round(vs, 4)
```

```
##      seed income_balance with_student
## [1,]    1         0.0254       0.0260
## [2,]    2         0.0238       0.0246
## [3,]    3         0.0264       0.0272
## [4,]    4         0.0256       0.0262
```

The four splits give validation error rates between 2.38% and 2.64%: the estimate varies with the split by about a quarter of a percentage point, Unit 1's first drawback of the validation set approach. Adding `student` raises the error slightly on every split (2.46% to 2.72%), so the dummy does not reduce the test error. ■

Exercise 7.30 (ISLR §5.4, exercise 6; beyond the syllabus)**.** For the logit of `default` on `income` and `balance`, compare the standard errors of `glm` with bootstrap standard errors.

Solution. The bootstrap is Unit 1's (beyond the syllabus):

```r
suppressMessages(library(boot))
gIB <- glm(default ~ income + balance, data = Df, family = binomial)
```

```r
bfun <- function(d, i) coef(glm(default ~ income + balance, data = d[i, ], family =
 ↪  binomial))[2:3]
set.seed(1); bt <- boot(Df, bfun, R = 1000)
signif(rbind(glm_se = coef(summary(gIB))[2:3, 2], bootstrap_se = apply(bt$t, 2,
 ↪  sd)), 4)
```

```
##                   income   balance
## glm_se       0.000004985 0.0002274
## bootstrap_se 0.000004866 0.0002299
```

The two agree to within 3% (4.99×10^{-6} against 4.87×10^{-6} for `income`, 2.27×10^{-4} against 2.30×10^{-4} for `balance`), as Theorem 7.19 leads one to expect when the logit is correctly specified: both estimate the sampling standard deviation of the estimate. ■

Exercise 7.31 (ISLR §5.4, exercise 7)**.** On `Weekly`: (a) fit the logit of `Direction` on `Lag1` and `Lag2`; (b) refit without the first observation; (c) classify the first observation; (d)–(e) compute the leave-one-out error with a loop.

Solution. LOOCV is Unit 1's, with the misclassification indicator as the loss (ISLR (5.4)):

```r
gL <- glm(Direction ~ Lag1 + Lag2, data = Wk, family = binomial); round(coef(gL), 4)
```

```
## (Intercept)        Lag1        Lag2
##      0.2212     -0.0387      0.0602
```

```r
g_1 <- glm(Direction ~ Lag1 + Lag2, data = Wk[-1, ], family = binomial)
c(p_up_first = round(predict(g_1, Wk[1, ], type = "response"), 4),
  actual_first = as.character(Wk$Direction[1]))
```

```
## p_up_first.1 actual_first
##     "0.5714"       "Down"
```

```r
loo <- sapply(seq_len(nrow(Wk)), function(i) {
  gi <- glm(Direction ~ Lag1 + Lag2, data = Wk[-i, ], family = binomial)
  cls(predict(gi, Wk[i, ], type = "response")) != Wk$Direction[i] })
round(c(loocv_error = mean(loo), always_up_error = mean(Wk$Direction != "Up")), 4)
```

```
##     loocv_error always_up_error
##          0.4500          0.4444
```

(a) The coefficients are 0.2212, −0.0387 and 0.0602. (c) Without it, the first week gets probability 0.5714 of `Up` and is classified `Up`; it was `Down`, an error. (e) The LOOCV error is 45.0%, against

44.4% for always predicting `Up`: the lags do not help. ■